

Micromagnetics Module User's Guide (V2.13)

Weichao Yu^{1,*}

¹*Institute for Nanoelectronic Devices and Quantum Computing, Fudan University*

(Dated: December 27, 2024)



* Email: wcyu@fudan.edu.cn

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A. Poster for CPS 2022, No. J8-P137 (Best Poster Prize)

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I. DEVELOPMENT TEAM

Main developers for the *Micromagnetics Module*:

Dr. Weichao Yu

Institute for Nanoelectronic devices and Quantum computing, Fudan University

Development of the Micromagnetics Module (Time Domain)

Packaging and distribution of the module file

Preparation of the manual and tutorial videos

Prof. Jiang Xiao

Department of Physics, Fudan University

Supervision of the project

Mr. Jiabin Zhang

Department of Physics, Fudan University

Development of the Micromagnetics Module (Frequency Domain)

Mr. Chensong Hua

Institute for Nanoelectronic devices and Quantum computing, Fudan University

Development of the Floquet Periodic Boundary Condition for Micromagnetics Module
(Frequency Domain)

Development of multiphysics coupling between RF module and Micromagnetics Module
(Frequency Domain)

Mr. Chongzhou Wang

Institute for Nanoelectronic devices and Quantum computing, Fudan University

Development of the magneto-elastic coupling approach in frequency domain

We acknowledge the help from **Mr. Bo Xiao** for preparing the **tutorial examples**.

II. STATEMENT

The *Micromagnetics Module* for COMSOL Multiphysics is mainly developed by Dr. Weichao Yu started from his PhD study under supervision of Prof. Jiang Xiao at Fudan University. The module is now being and will be further updated and maintained at *Institute for Nanoelectronic devices and Quantum computing, Fudan University*.

The *Micromagnetics Module* is designed for COMSOL users who are researchers, engineers and students working in the field of Magnonics or Spintronics, and any other fields which is related to magnetism. The users are supposed to (i) be familiar with the operation of COMSOL Multiphysics and (ii) have academic background on condensed-matter physics/magnetism/spintronics/magnonics. Required basic knowledge will NOT be introduced in this user's guide. If you are a beginner, the following materials are suggested for you:

For beginners on COMSOL Multiphysics, you can learn basically everything from the Reference Manual. You can find it at (if you install COMSOL to Windows in a default way)

```
C:\Program Files\COMSOL\COMSOLXX\Multiphysics\doc
\pdf\COMSOL_Multiphysics\COMSOL_ReferenceManual.pdf
```

where "XX" is the version of COMSOL, such as 56 for version 5.6. You can also visit the official site <https://comsol.com> for other learning materials.

For beginners on magnonics, it's recommended to read Dr. Weichao Yu's [PhD thesis](#) (written in Chinese). The *Micromagnetics Module* is developed following the roadmap in the thesis. Non-Chinese users can check the English publications listed in the end and the [review article](#) which cover most of the contents in the thesis.

While using the *Micromagnetics Module*, please follow the regulations of COMSOL company as well as the laws of your country. Do not crack the module file by yourself. For any questions and bugs, please report via email to wcyu@fudan.edu.cn. If not stated, the examples of this user's guide are conducted using COMSOL v5.6 and the module file may not be compatible with lower version.

Announcements and update of the *Micromagnetics Module* can be found at [Application Exchange](#) on COMSOL official website. You are encouraged to have a read on COMSOL Blog ([Chinese/English](#)). Tutorial videos with [examples](#) are updated on [Bilibili](#) (only in Chinese).

III. INSTALL AND UPDATE

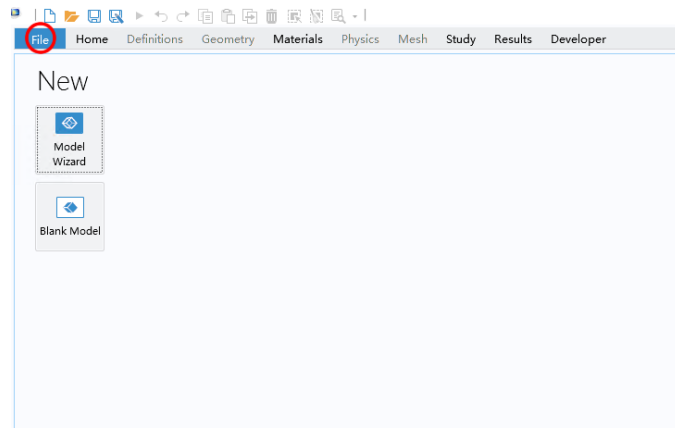
How to install. When you are authorized to become a user of the *Micromagnetics Module*, you will be distributed with a single file named as `Micromagnetics Module.jar`.

Step 1. Copy the `.jar` file to the COMSOL installation folder (You may need authorization from the administrator.). For Windows users, the default path is

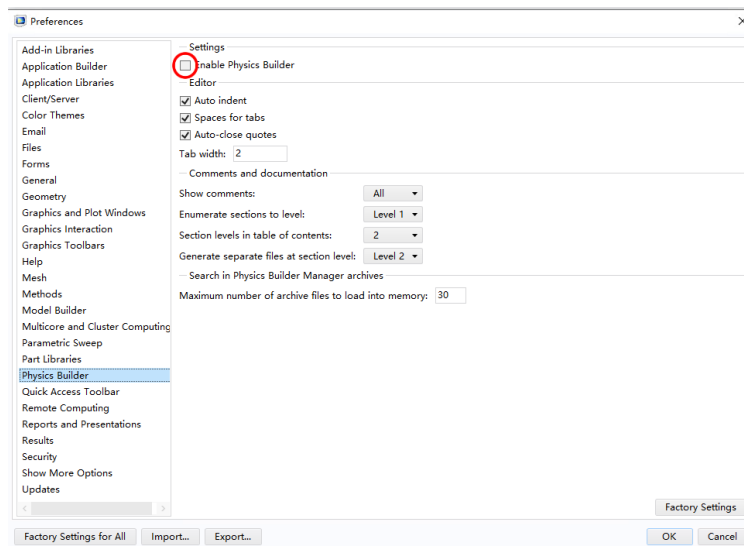
`C:\Program Files\COMSOL\COMSOL56\Multiphysics\plugins`

The `.jar` file will be automatically imported when the following setup is finished.

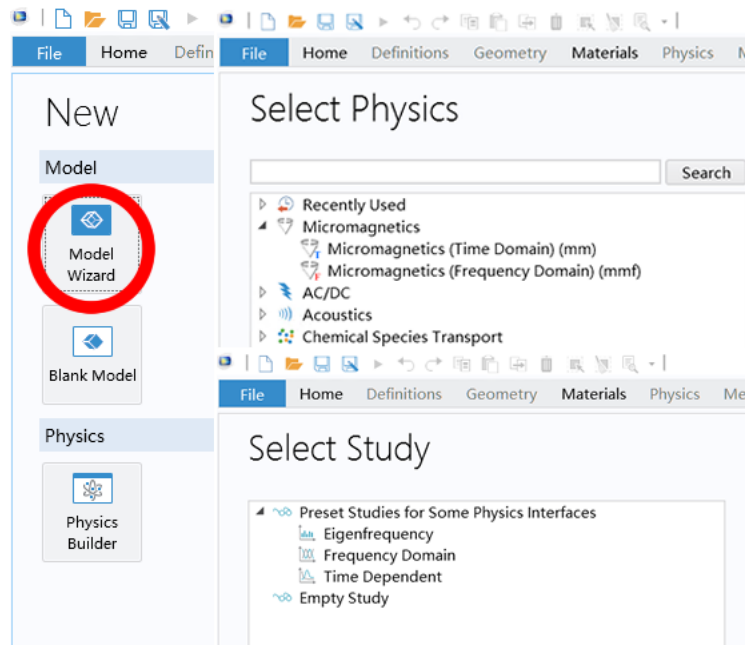
Step 2. Run COMSOL Multiphysics, and you will see the starting interface for default installation as shown below, which indicates that the **Physics Builder** is not enabled yet.



Step 3. Click **File\Preferences\Physics Builder** and click the checkbox to enable Physics Builder. Click **OK** and restart COMSOL Multiphysics.



Step 4. Restarting COMSOL Multiphysics, you will see an extra icon, indicating that now you can import customized physics. Click **Model Wizard** and select physics. Now you can see a new option, which is the *Micromagnetics Module* we imported. Skip the first three steps if the **Physics Builder** is already enabled.



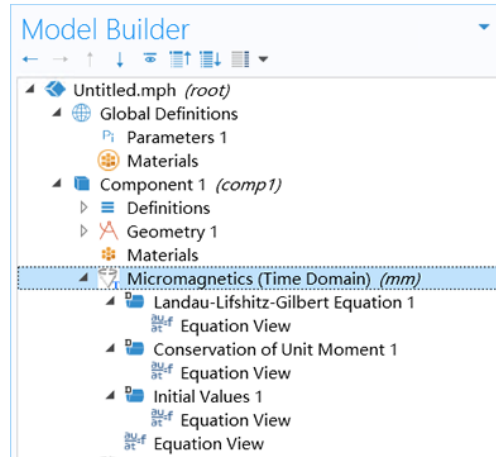
Since Version 2.0, the *Micromagnetics Module* includes two parts, one is the *Micromagnetics (Time Domain)* which solves the time-dependent Landau-Lifshitz-Gilbert equation, thus supporting **Time Dependent** study, the other one is the *Micromagnetics (Frequency Domain)* which solves the linearized Landau-Lifshitz-Gilbert equation, thus supporting **Frequency Domain** and **Eigenfrequency** study.

How to update. The functions of *Micromagnetics Module* will be further updated in the future. When you get the updated module file (which will be named same as the original file), copy it to the same path mentioned above and overwrite the original file.

Warning: Make sure that the module file name is "Micromagnetics Module.jar", otherwise there will be errors when initializing the software.

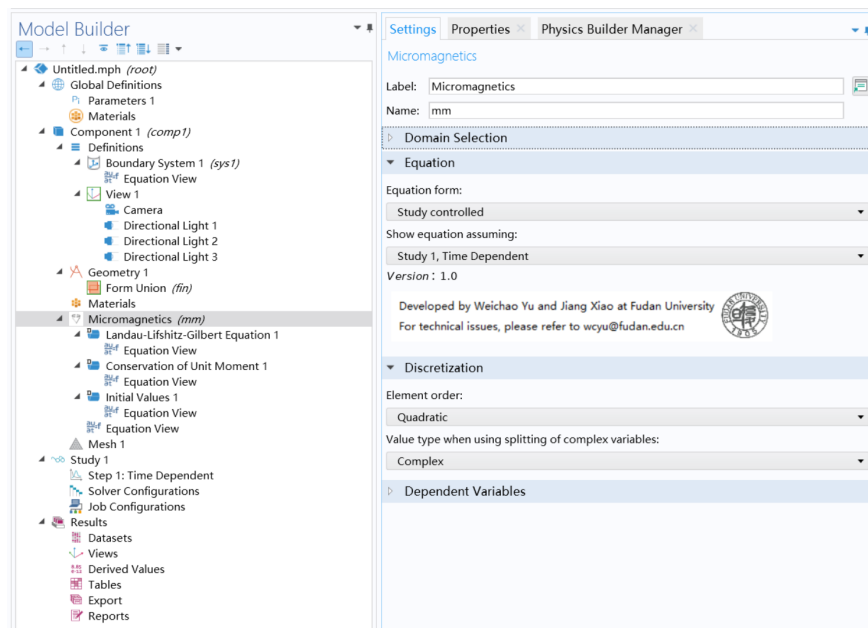
IV. MODELLING WITH MICROMAGNETICS MODULE (TIME DOMAIN)

Now we can perform time-dependent simulation using *Micromagnetics Module (Time Domain)* in the same way as other preset modules in COMSOL.



A. Main Node: Micromagnetics (Time Domain) (mm)

Here the dependent variables are defined. It's well known that the magnetic moment is modeled as a three-dimensional vector $\mathbf{m} = (m_x, m_y, m_z)$ and they are defined in COMSOL as mX , mY , mZ . The version number will be displayed at the main node.



B. Governing equation: Landau-Lifshitz-Gilbert equation

1. Equation Overview

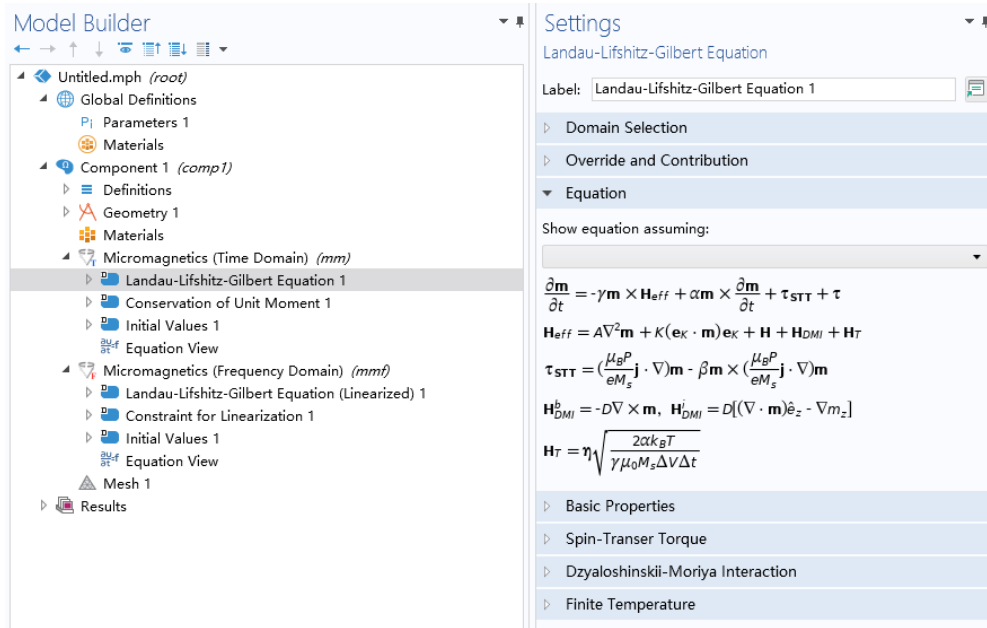
The governing equation for micromagnetics is Landau-Lifshitz-Gilbert equation:

$$\frac{\partial \mathbf{m}}{\partial t} = -\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}} + \alpha \mathbf{m} \times \frac{\partial \mathbf{m}}{\partial t} + \boldsymbol{\tau}_{\text{STT}} + \boldsymbol{\tau}, \quad (1)$$

where $\mathbf{m}$ is the macrospin vector and $\mathbf{H}_{\text{eff}}$ is the effective field whose components vary with materials. $\boldsymbol{\tau}_{\text{STT}}$ is spin-transfer torque and $\boldsymbol{\tau}$ is other form of torques such as spin-orbit torque which can be customized by the user. We consider the effective field as

$$\mathbf{H}_{\text{eff}} = A \nabla^2 \mathbf{m} + K(\mathbf{e}_K \cdot \mathbf{m})\mathbf{e}_K + \mathbf{H} + \mathbf{H}_{\text{DMI}} + \mathbf{H}_T, \quad (2)$$

which includes exchange interaction, uniaxial anisotropy, external field, Dzyaloshinskii-Moriya interaction (DMI) and random field induced by thermal fluctuations.



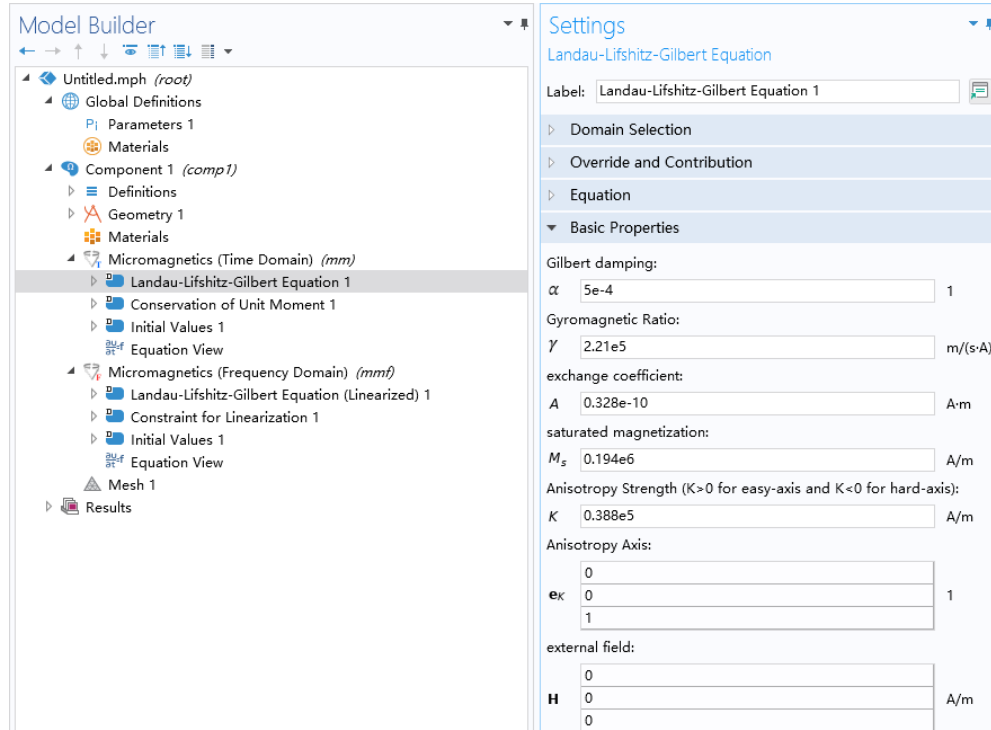
By expanding the following sections, one can input parameters and customize effective fields and torques.

2. Basic Properties

Here you can input basic parameters, such as **Gilbert damping**, **gyromagnetic ratio**, **exchange coefficient** and **saturated magnetization**. The default parameters are chosen for YIG.

The anisotropy field is characterized by the scalar **anisotropy strength** K and the vector **anisotropy axis** $\mathbf{e}_K$. A positive K indicates for the easy-axis anisotropy while a negative one indicates for the hard-axis anisotropy (or easy-plane normal to the hard axis). Note that $\mathbf{e}_K$ doesn't have to be normalized since it will be normalized automatically by the code. The **external field** $\mathbf{H}$ is initially designed for static fields. However, it can be used in a more powerful way. For example, input $100 [\text{A/m}] * \sin(2 * \pi * 1 [\text{GHz}] * t)$ to apply an oscillating field with certain amplitude and frequency, or input $t_{\text{emw}}.H_x$ *etc.* to call solutions from other physical interfaces such as **RF** module.

Other than **external field** or **uniaxial anisotropy**, the remaining parameters mentioned above are all **Variables**, which means they can be a function of space, time and other variables. For example, input $5e-4 * (\text{dom}==1) * (t < 1 [\text{ns}])$ indicating that the damping at region **Domain 1** is 5×10^{-4} before $t = 1 \text{ ns}$ and is zero after that, and for the remaining regions the damping is always zero.

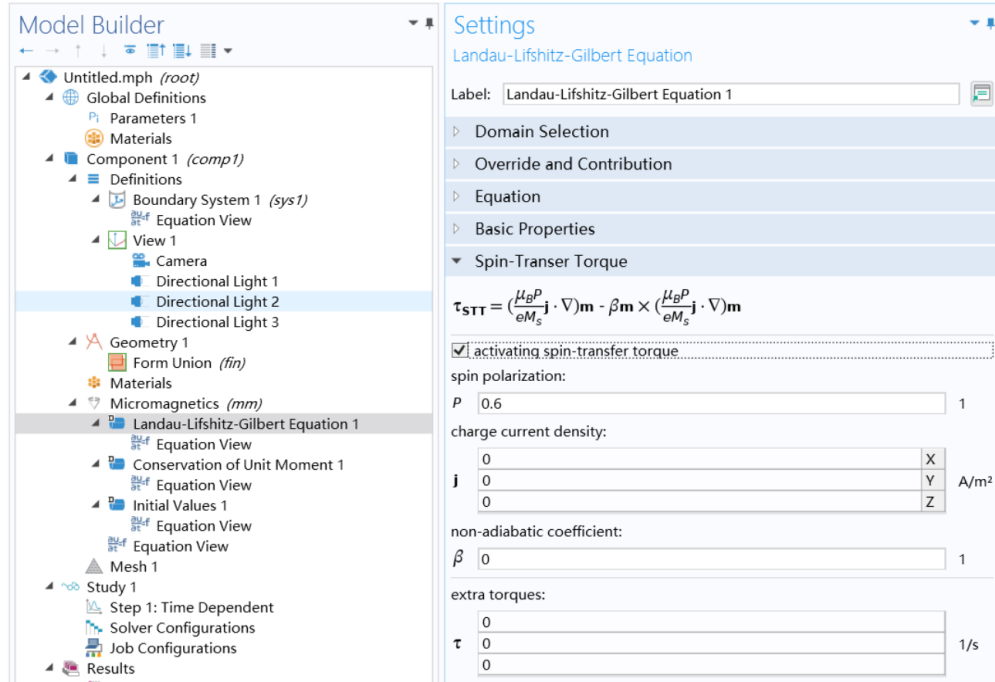


3. Spin-Transfer Torque

The **spin-transfer torque** (STT) terms read as

$$\boldsymbol{\tau}_{\text{STT}} = \left(\frac{\mu_B P}{e M_s} \mathbf{j} \cdot \nabla \right) \mathbf{m} - \beta \mathbf{m} \times \left(\frac{\mu_B P}{e M_s} \mathbf{j} \cdot \nabla \right) \mathbf{m}, \quad (3)$$

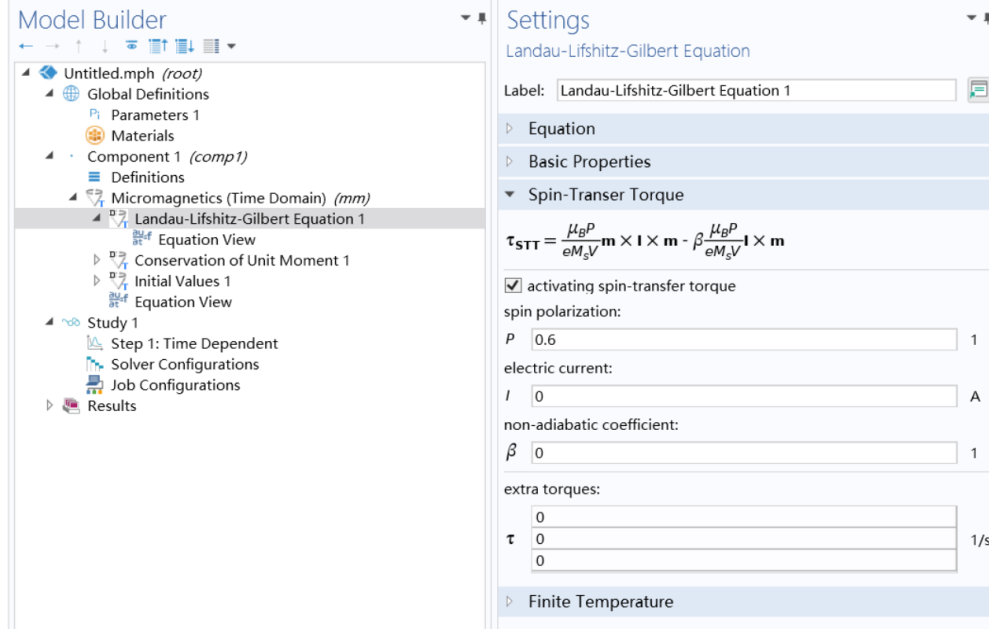
where μ_B is Bohr magneton and e is electron charge, whose values are embedded in the module. Users can input **spin polarization factor** P , **charge current density** and **non-adiabatic coefficient** β . Here, the **charge current density** is a vector having the same dimension with the model, which can be input manually or coupled to other physical interfaces, such as *AC/DC Module Electric Currents* to be introduced later. Check the **activating spin-transfer torque** box to active the STT terms and clear it when STT is absent.



Since V2.1, the *Micromagnetics Module (Time Domain)* is able to perform simulation in 0D. Since there is no spatial dependence of the dependent variable, the spin-transfer torque term is modified as

$$\tau_{\text{STT}} = \frac{\mu_B P}{e M_s V} \mathbf{m} \times \mathbf{I} \times \mathbf{m} - \beta \frac{\mu_B P}{e M_s V} \mathbf{I} \times \mathbf{m}, \quad (4)$$

with electric current $\mathbf{I}$ and realistic volume V of the magnet (although it's modelled as a point in 0D simulation). It should be noticed that the direction of electric current is presumed to be **along x direction** so that one only needs to input a scalar value in the interface. This form is widely used to simulate a magnetic tunnel junction (MTJ) (See [Camsari et al., Implementing p-bits With Embedded MTJ](#)).



4. Dzyaloshinskii-Moriya Interaction

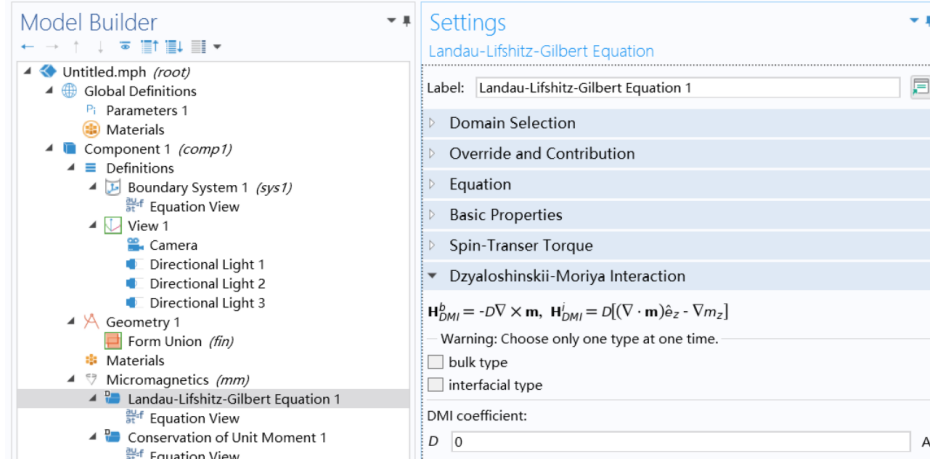
There are two kinds of **Dzyaloshinskii-Moriya Interaction**, the bulk-type one and interfacial-type one:

$$\mathbf{H}_{\text{DMI}}^b = -D \nabla \times \mathbf{m}, \quad \mathbf{H}_{\text{DMI}}^i = D[(\nabla \cdot \mathbf{m})\hat{e}_z - \nabla m_z]. \quad (5)$$

You can choose either type by clicking the checkbox. You can also input the strength of DMI, *i.e.* **DMI coefficient** below, whose default value is $D = 0$ A, indicating the absence of DMI. You can also turn off DMI by clearing both checkboxes.

Warning: You are supposed to choose only one of the two types at one time, otherwise the physics is not correct.

It should be noticed that for the interfacial-type DMI, we assume the normal vector of the interface is along z direction. In the presence of DMI, appropriate boundary conditions should also be applied. The boundary conditions will be introduced in Section IV E.

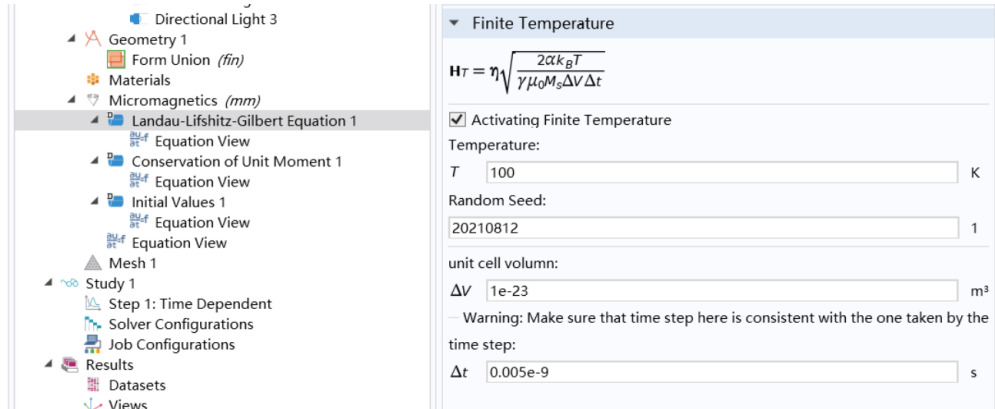


5. Finite Temperature

In the presence of finite temperature, thermal fluctuations are modeled by random effective fields, which takes the following form

$$\mathbf{H}_T = \boldsymbol{\eta} \sqrt{\frac{2\alpha k_B T}{\gamma \mu_0 M_s \Delta V \Delta t}}, \quad (6)$$

with k_B the Boltzmann constant, μ_0 the vacuum permeability and T the temperature to be input in Kelvin scale.



The randomness of the thermal fluctuation field is introduced by a vector random function $\boldsymbol{\eta}$ with normal distribution (mean is 0 and standard deviation is 1). The random function follows the restriction that $\langle \eta^i(t) \eta^j(t') \rangle = \delta_{ij} \delta(t - t')$ with $i, j = x, y, z$, indicating that there is no spatial and temporal correlation for the random function. You can input any positive integer in the field of **Random Seed** to generate different sequence of random number in each simulation, *e.g.* you can input the date today. ΔV is the volume of a unit cell and can be replaced by the mesh volume

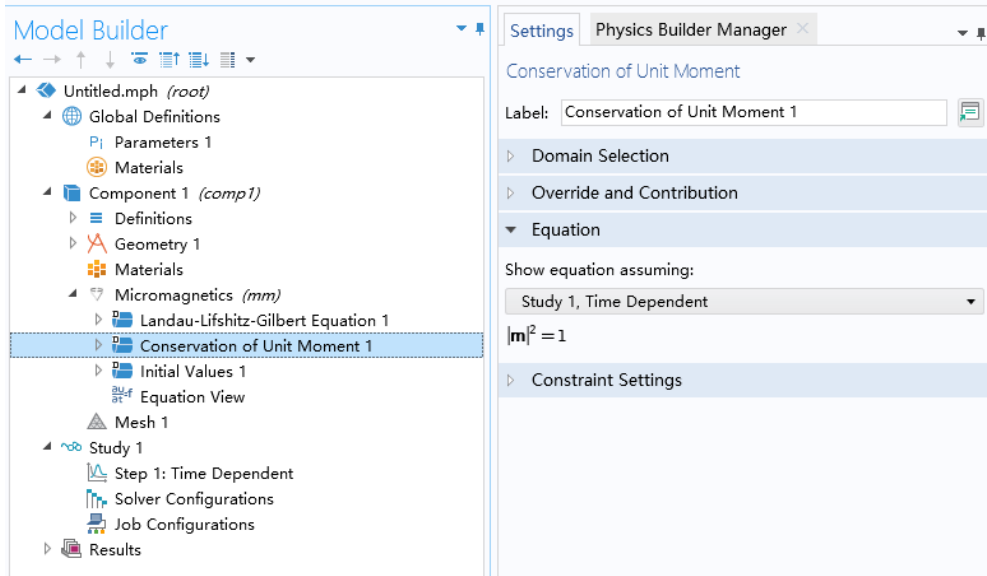
in finite elements method. In general cases, you can input a constant value such as 10^{-22} m^3 . For extremely inhomogeneous meshes, you can use the built-in variable `meshvol` to evaluate the mesh size. Δt is the time step taken in your simulation. Check the **activating finite temperature** box to activate the thermal effects and clear it for zero temperature (equivalent to setting $T = 0 \text{ K}$).

Warning: Make sure that time step here is consistent with the one taken in the solver, otherwise the energy distribution of magnon excitation will be incorrect, equivalent to a wrong estimation of temperature.

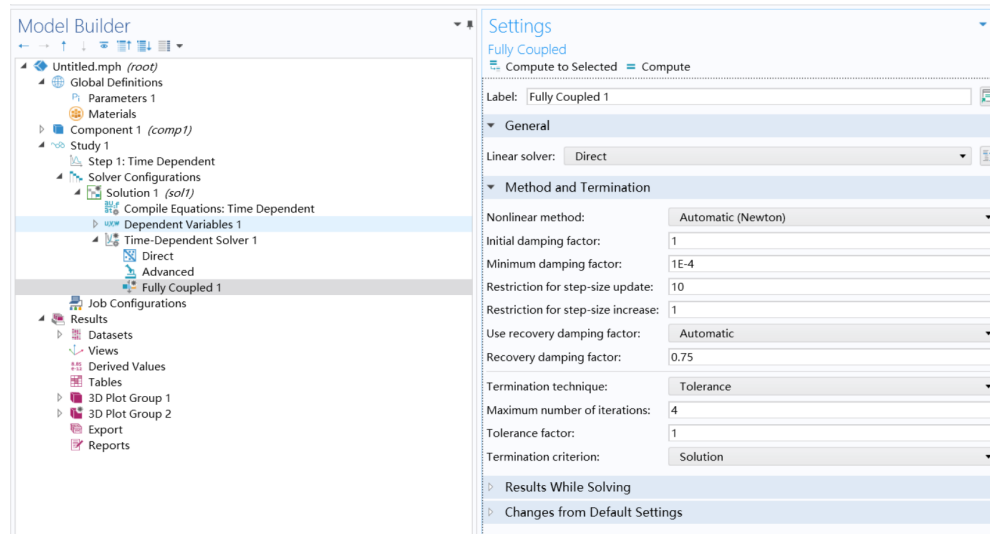
C. Conservation of Unit Moment

The micromagnetic model requires that the magnitude of the magnetic moment should be conserved:

$$m_x^2 + m_y^2 + m_z^2 = 1 \quad \text{or} \quad |\mathbf{m}|^2 = 1. \quad (7)$$



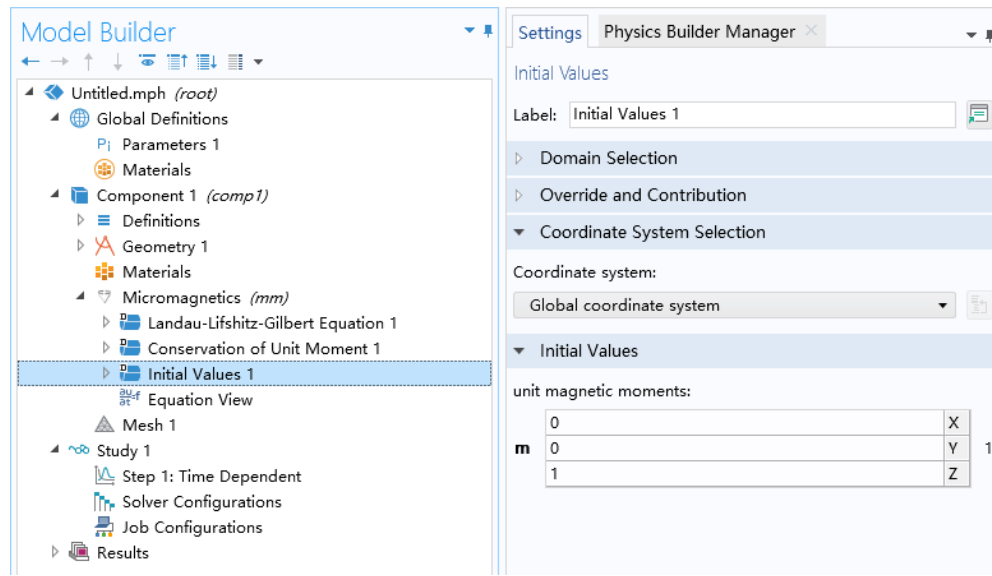
Such a constraint is necessary for LLG equation expressed in **Cartesian coordinate**, but automatically fulfilled when the LLG equation is expressed in **spherical coordinate** (m_x, m_y, m_z) = $(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$. Solving LLG equation in **spherical coordinate** (with only two dependent variables θ and ϕ) can accelerate the computation, but suffers from singularity problems. The *Micromagnetics Module* in **spherical coordinate** is scheduled to be updated in the future version.



Warning: Make sure that the nonlinear method taken by the Time-dependent Solver is always **Automatic (Newton)**, since we have applied a global constraint on the magnitude of m , otherwise the simulation may not converge.

D. Initial Value

To solve PDE, initial values for dependent variables are required. The default initial state for the magnetization is set to along z direction. It's suggested to set **anisotropy** or **external field** parallel with the initial magnetization to avoid numerical failures. The initial values here are also variables, which can be spatially dependent, so that initial profile of domain wall, skyrmion and other magnetic textures can be input here. Except defining initial values by manually inputting explicit expressions, users can define interpolating function based on external data.

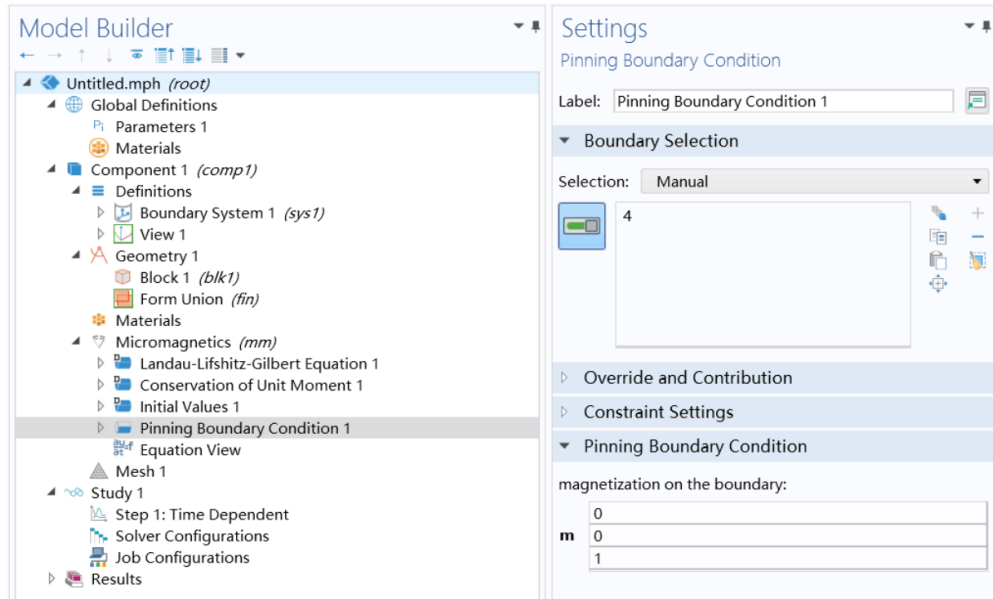
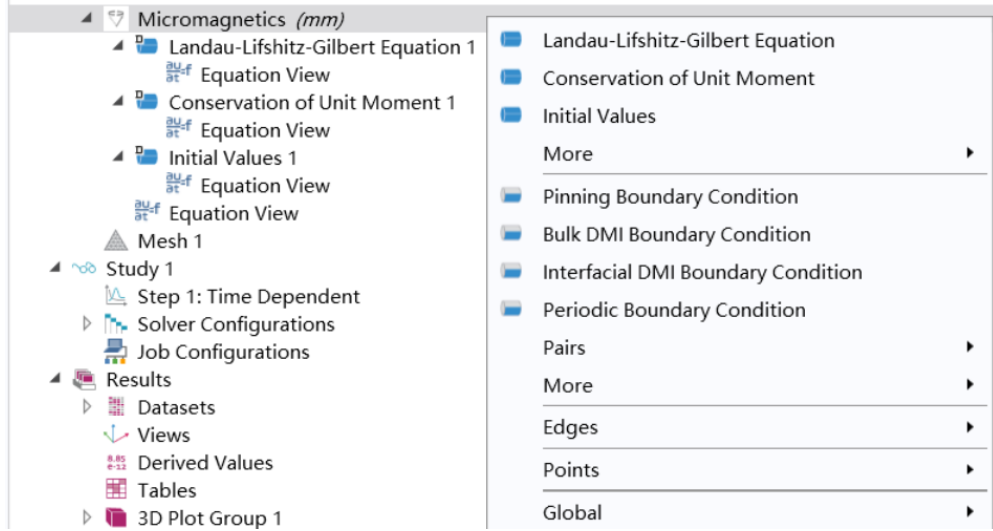


E. Optional Features

Right-clicking the main node, you will get extra options to facilitate your modelling.

1. Pinning Boundary Condition

When you apply **Pinning Boundary Condition**, you are forcing the value of magnetization on chosen boundaries to be fixed. For example, input $\mathbf{m} = \{0, 0, 1\}$ to force the magnetization on the chosen boundaries to align in z direction. Remember to keep the norm of $\mathbf{m}$ to be 1, otherwise numerical problems may occur.

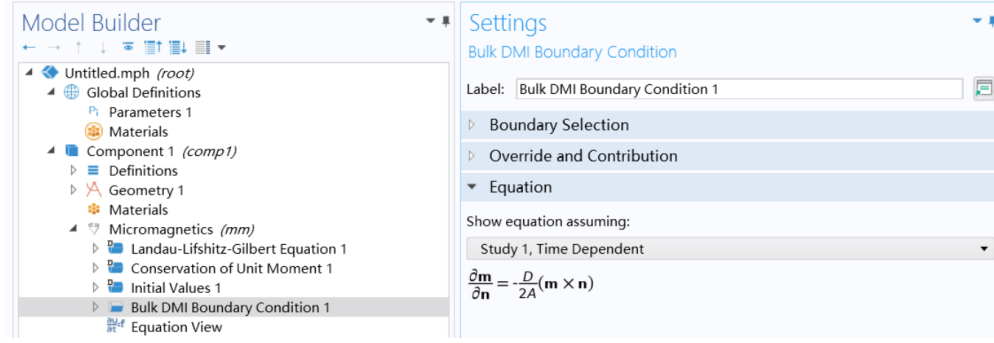


2. Bulk DMI Boundary Condition

In the presence of bulk-type DMI, remember to apply **Bulk DMI Boundary Condition** with following expression:

$$\frac{\partial \mathbf{m}}{\partial \mathbf{n}} = -\frac{D}{2A}(\mathbf{m} \times \mathbf{n}). \quad (8)$$

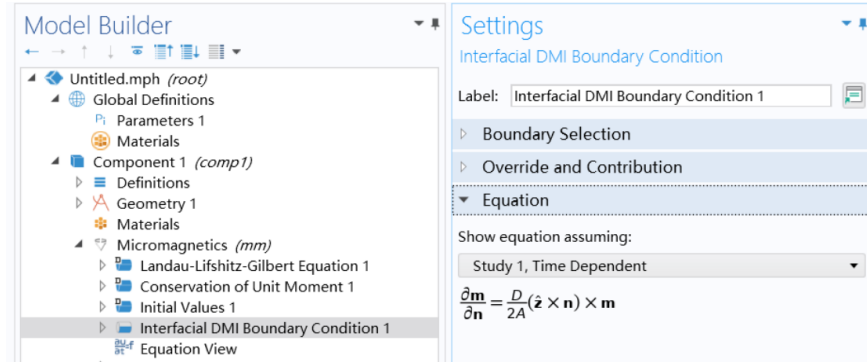
Remember to check the **bulk type** checkbox in Section.IV B 4. This boundary condition works for 1D, 2D and 3D modelling.



3. Interfacial DMI Boundary Condition

In the presence of interfacial-type DMI, remember to apply **Interfacial DMI Boundary Condition** with following expression:

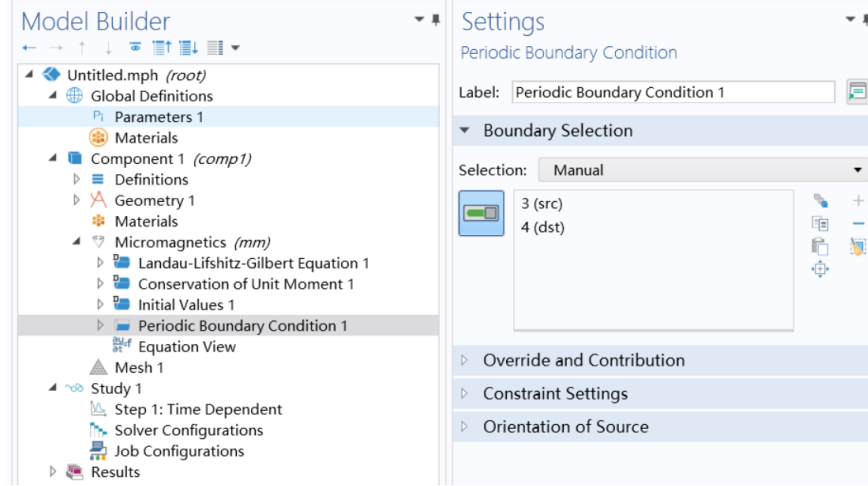
$$\frac{\partial \mathbf{m}}{\partial \mathbf{n}} = \frac{D}{2A}(\hat{\mathbf{z}} \times \mathbf{n}) \times \mathbf{m}. \quad (9)$$



Remember to check the **interfacial type** checkbox in Section.IV B 4. Due to the broken inversion symmetry of interfacial-type DMI and assumption that normal vector of interface is along z direction, the boundary condition only takes effect on boundaries in $x - y$ plane.

4. Periodic Boundary Condition

In order to simulate models with periodic features such as magnonic crystals, periodic boundary condition is needed. Choose boundaries, and system will automatically recognize them as source (src) and destination (dst).

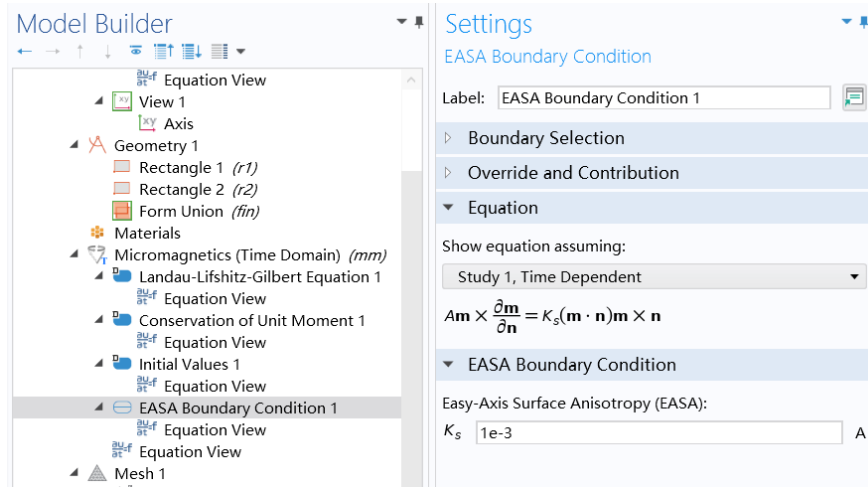


5. Easy-Axis Surface Anisotropy (EASA) Boundary Condition

This is a new type of boundary condition included into the module since V2.03. The function of the EASA boundary condition is to simulate materials with certain surface anisotropy, whose easy-axis is along the norm of the surface. The boundary condition is expressed analytically as

$$A\mathbf{m} \times \frac{\partial \mathbf{m}}{\partial \mathbf{n}} = K_s(\mathbf{m} \cdot \mathbf{n})\mathbf{m} \times \mathbf{n}, \quad (10)$$

with K_s the EASA strength whose dimension is A (Ampere).



The presence of EASA can induce surface exchange spin waves and one can use such spatially modulated surface to design **spin-wave circuits**. For details of the derivation, please refer to **the original work**. This boundary condition works for 2D and 3D modelling.

V. MODELLING WITH MICROMAGNETICS MODULE (FREQUENCY DOMAIN)

A unique feature (since V2.0) of the *Micromagnetics Module* is its ability to perform micromagnetic simulation in frequency domain, which is efficient when studying eigenmodes for arbitrary magnetic textures or excitation under monochromatic perturbation.

A. Theoretical Preparation: Linearized LLG Equation in Frequency Domain

It's straightforward to solve time-dependent LLG equation in time domain. However in frequency domain, we first need to perform Fourier transformation, *i.e.*, $\mathbf{m}(\mathbf{r}, t) \rightarrow \mathbf{m}(\mathbf{r}, \omega)$, with ω the angular frequency. Starting from the time-dependent Landau-Lifshitz-Gilbert equation Eq.(1) without torque terms

$$\frac{\partial \mathbf{m}}{\partial t} = -\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}} + \alpha \mathbf{m} \times \frac{\partial \mathbf{m}}{\partial t}. \quad (11)$$

Assuming that the magnetic moment $\mathbf{m}$ is perturbed around its equilibrium magnetization $\mathbf{m}_0$ with certain angular frequency ω , it can be decomposed as $\mathbf{m} = \mathbf{m}_0 + \delta \mathbf{m} e^{i\omega t}$ (note that we follow the convention $e^{i\omega t}$ rather than $e^{-i\omega t}$) under linear approximation ($\delta \mathbf{m} \ll \mathbf{m}_0$). Another approximation $\mathbf{m}_0 \cdot \delta \mathbf{m} = 0$ is also considered, indicating that the excitation is always orthogonal to the static component, so that the magnitude of the static magnetic moment can still be treated as a unit vector. Following the same principle, the effective field can be decomposed into two parts as $\mathbf{H}_{\text{eff}} = \mathbf{h}_0^{\text{eff}} + \delta \mathbf{h}^{\text{eff}} e^{i\omega t}$, with the static component and the dynamical one. Substituting $\mathbf{m}$ and $\mathbf{H}_{\text{eff}}$ back to the time-dependent LLG equation, we obtain

$$i\omega \delta \mathbf{m} e^{i\omega t} = -\gamma \mathbf{m}_0 \times \delta \mathbf{h}^{\text{eff}} e^{i\omega t} - \gamma \delta \mathbf{m} e^{i\omega t} \times \mathbf{h}_0^{\text{eff}} + i\omega \alpha \mathbf{m}_0 \times \delta \mathbf{m} e^{i\omega t}. \quad (12)$$

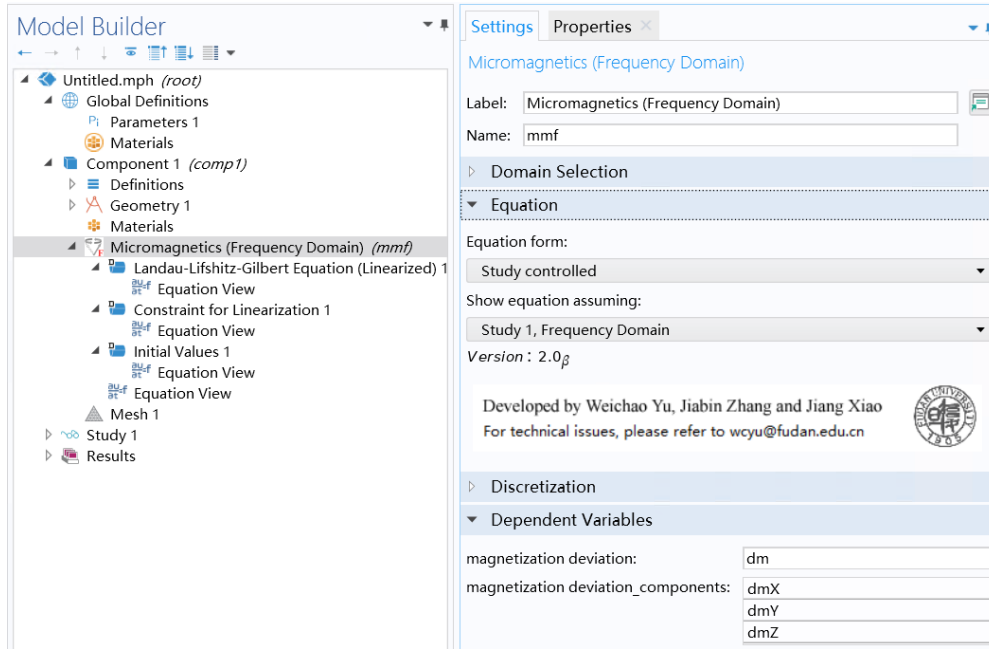
Notice that some terms are discarded during derivation, *e.g.*, $\partial \mathbf{m}_0 / \partial t = 0$ since $\mathbf{m}_0$ is the static component, $\mathbf{m}_0 \times \mathbf{h}_0^{\text{eff}} = 0$ since the magnetization in equilibrium should be parallel with the static effective field, and $\delta \mathbf{m} e^{i\omega t} \times \delta \mathbf{h}^{\text{eff}} e^{i\omega t} \sim e^{2i\omega t}$ which is in high order. Eliminating the same harmonic factor $e^{i\omega t}$ on both sides of the equation, we obtain the linearized LLG equation in frequency domain,

$$i\omega \delta \mathbf{m} = -\gamma \mathbf{m}_0 \times \delta \mathbf{h}^{\text{eff}} - \gamma \delta \mathbf{m} \times \mathbf{h}_0^{\text{eff}} + i\omega \alpha \mathbf{m}_0 \times \delta \mathbf{m}, \quad (13)$$

which is a stationary partial differential equation (for given angular frequency ω) to be solved in the *Micromagnetics Module (Frequency Domain)*.

B. Main Node: Micromagnetics (Frequency Domain) (mmf)

Different from the dependent variables $\mathbf{m} = (m_x, m_y, m_z)$ in *Micromagnetics Module (Time Domain)* which are real numbers, here in *Micromagnetics Module (Frequency Domain)*, the dependent variables are $\delta\mathbf{m} = (\delta m_x, \delta m_y, \delta m_z)$, which are complex numbers, or complex amplitudes (phasors). The three components are named as dmX , dmY , dmZ by default in the module. The solution in frequency domain follows phasor arithmetic rules, and can be transformed back to time domain according to $\mathbf{m}(t) = \mathbf{m}_0 + \text{Re}(\delta\mathbf{m}e^{i\omega t})$. Taking one of the scalar components δm_x as example, the complex solution can always be decomposed as $\delta m_x = A(\cos \varphi + i \sin \varphi) = Ae^{i\varphi}$, with A the absolute value of δm . Hence the solution mapped back to time domain can be written as $\text{Re}(\delta m_x e^{i\omega t}) = \text{Re}(Ae^{i\varphi} e^{i\omega t}) = \text{Re}(Ae^{i(\omega t + \varphi)}) = A \cos(\omega t + \varphi)$, where the information of phase is encoded into the relative magnitude between real and imaginary parts.



1. Basic Properties

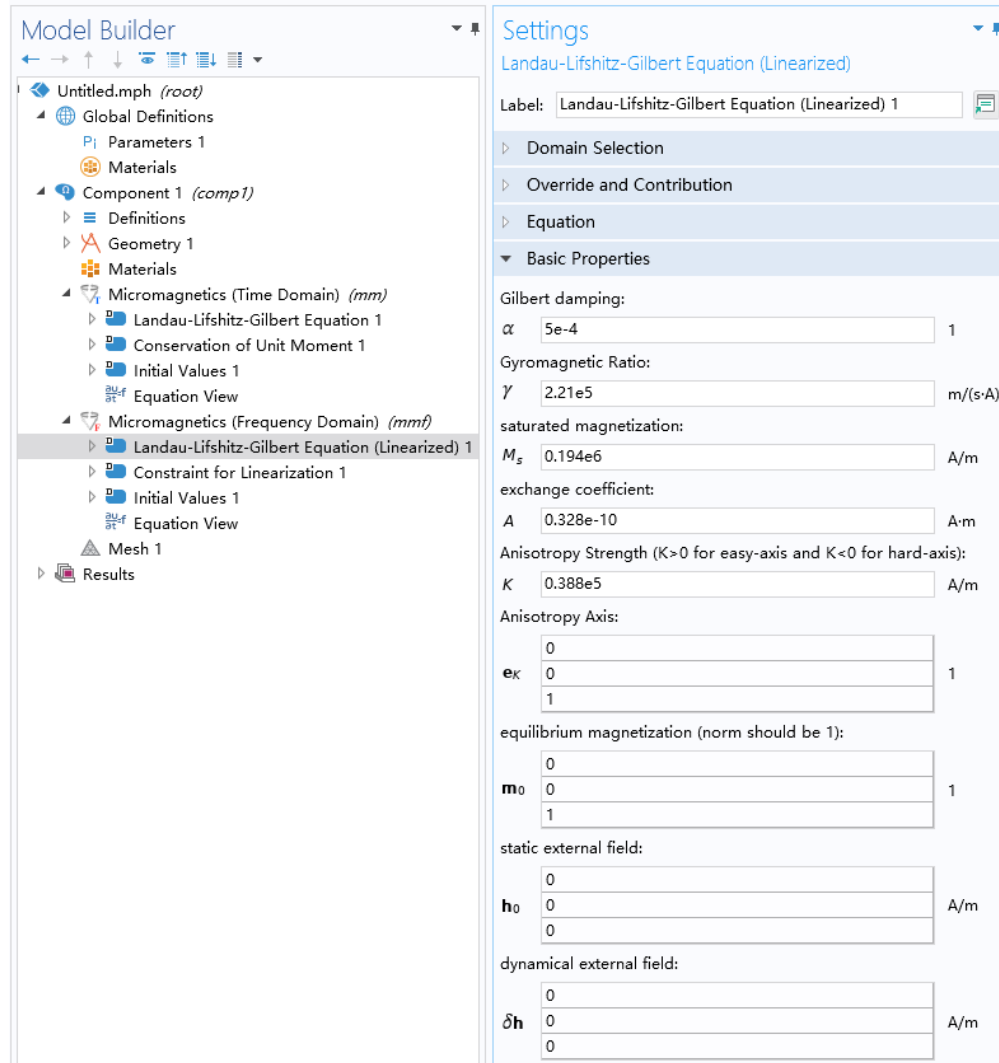
The interface for basic properties is similar to the one of *Micromagnetics Module (Time Domain)*. Following the same procedure of deriving linearized LLG equation, the effective field including exchange interaction, uniaxial anisotropy and external field can be decomposed into two parts, with static component

$$\mathbf{h}_0^{\text{eff}} = A\nabla^2\mathbf{m}_0 + K(\mathbf{e}_K \cdot \mathbf{m}_0)\mathbf{e}_K + \mathbf{h}_0, \quad (14)$$

and the dynamical component

$$\delta \mathbf{h}^{\text{eff}} = A \nabla^2 \delta \mathbf{m} + K(\mathbf{e}_K \cdot \delta \mathbf{m})\mathbf{e}_K + \delta \mathbf{h}. \quad (15)$$

The effective fields of exchange interaction and uniaxial anisotropy have been built into the module, while both the **static external field** $\mathbf{h}_0$ and the **dynamical external field** $\delta \mathbf{h}$ can be customized by the user (zero by default). It should be noticed that when defining **dynamical external field** $\delta \mathbf{h}$, just input the magnitude of the dynamical field *without* following sinusoidal term such as $\cos(\omega t + \varphi)$. In the *Micromagnetics Module (Frequency Domain)*, the harmonic factor $e^{i\omega t}$ with angular frequency ω (defined by the solver) is automatically attached to the dynamical components.



Another new interface is the **Equilibrium Magnetization** $\mathbf{m}_0$, which should be a unit vector. Users are supposed to define the correct equilibrium magnetization distribution, which should be

ground state, or at least stable state for given effective fields. Otherwise the solution may not be physical even if it can be successfully solved by COMSOL. In the simplest case (see figure above), the magnetization is uniformly distributed in $+z$ direction, consistent with the uniaxial anisotropy (along z axis). In the presence of complex magnetic textures, it's always difficult to determine the texture profile by explicit expressions, but it's easy to do numerically with the help of *Micromagnetics Module (Time Domain)*, where a stable magnetization distribution can be obtained by solving time-dependent LLG equation with large damping. The method of importing solutions from *Micromagnetics Module (Time Domain)* will be introduced later with an example (Section [V G](#)).

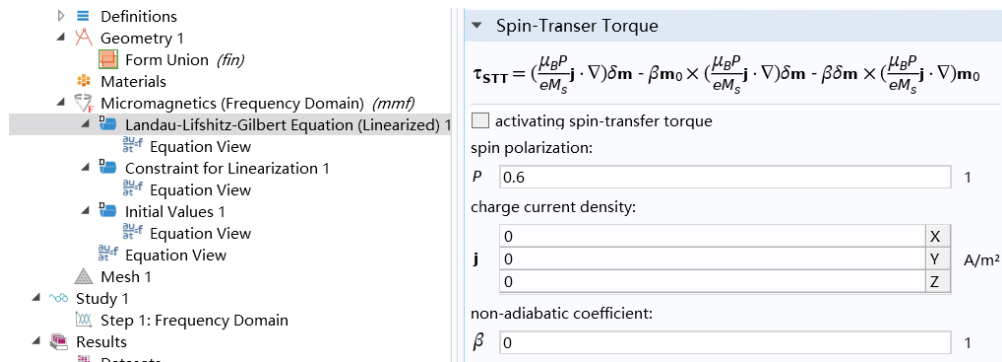
Note: Remember that magnitude of $\mathbf{m}_0$ is 1, so make sure that the magnitude of $\delta\mathbf{m}$ is much smaller than one. Otherwise the results will not be physical.

2. Spin-Transfer Torque

Assuming the electric current is constant with time, the **spin-transfer torque** (STT) terms can be decomposed into static and dynamical parts,

$$\tau_{\text{STT}} = \left(\frac{\mu_B P}{e M_s} \mathbf{j} \cdot \nabla \right) \delta \mathbf{m} - \beta \mathbf{m}_0 \times \left(\frac{\mu_B P}{e M_s} \mathbf{j} \cdot \nabla \right) \delta \mathbf{m} - \beta \delta \mathbf{m} \times \left(\frac{\mu_B P}{e M_s} \mathbf{j} \cdot \nabla \right) \mathbf{m}_0, \quad (16)$$

where one of the adiabatic terms $\sim \mathbf{m}_0$ and two of the non-adiabatic terms $\sim \mathbf{m}_0 \times \mathbf{m}_0$ and $\delta \mathbf{m} \times \delta \mathbf{m}$ are dropped.



The effect of spin-transfer torque can be turned on by checking the **Activating Spin-Transfer Torque** box. Here the **Charge Current Density** are assumed to be static components which doesn't vary with time.

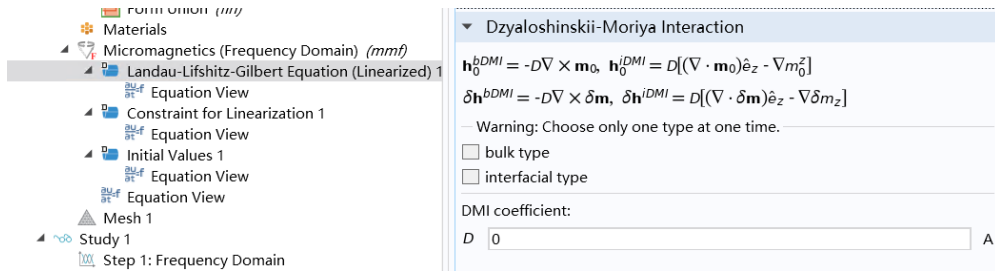
3. Dzyaloshinskii-Moriya Interaction

The bulk- and interfacial-type of **Dzyaloshinskii-Moriya Interaction** can be decomposed into a static component

$$\mathbf{h}_0^{\text{bDMI}} = -D\nabla \times \mathbf{m}_0, \quad \mathbf{h}_0^{\text{iDMI}} = D[(\nabla \cdot \mathbf{m}_0)\hat{e}_z - \nabla m_0^z], \quad (17)$$

and a dynamical component

$$\delta\mathbf{h}^{\text{bDMI}} = -D\nabla \times \delta\mathbf{m}, \quad \delta\mathbf{h}^{\text{iDMI}} = D[(\nabla \cdot \delta\mathbf{m})\hat{e}_z - \nabla \delta m_z]. \quad (18)$$



The two components are already built in the module, and the users needs to decide which type to activate and how large the **DMI coefficient** should be. The issues to be noticed are same as the one in *Micromagnetis Module (Time Domain)* (see Section IV B 4).

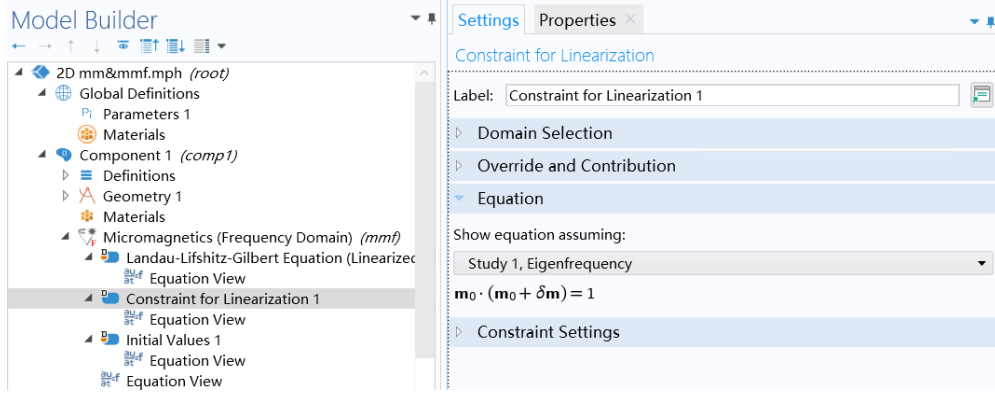
Note: It's still unknown how the DMI boundary condition can be correctly modelled.

C. Condition for Linearization

Dependent variables in the *Micromagnetics Module (Frequency Domain)* are dynamical excitations $\delta\mathbf{m}$, whose magnitude is proportional to external perturbation, so the constraint on the magnitude is not necessary. However, to keep the magnitude of equilibrium magnetization $\mathbf{m}_0$ conserved, we apply the approximation that the dynamical excitation is always orthogonal to the equilibrium magnetization,

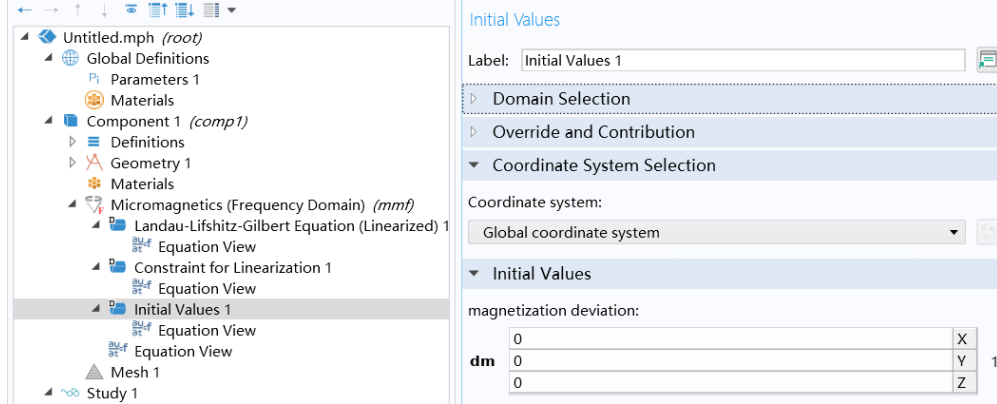
$$\mathbf{m}_0 \cdot (\mathbf{m}_0 + \delta\mathbf{m}) = 1, \quad (19)$$

where $\mathbf{m}_0 \cdot \mathbf{m}_0 = 1$ and $\mathbf{m}_0 \cdot \delta\mathbf{m} = 0$. The constraint is applied globally in the simulation.



D. Initial Value

The default initial values for the dependent variables are $\delta\mathbf{m} = \{0, 0, 0\}$, corresponding to a static state, which is usually an appropriate choice for **Eigenfrequency** study. For **Frequency Domain** study, an external perturbation $\delta\mathbf{h}$ needs to be applied according to realistic experimental setup, otherwise the system will not be excited and the solution is zero everywhere.

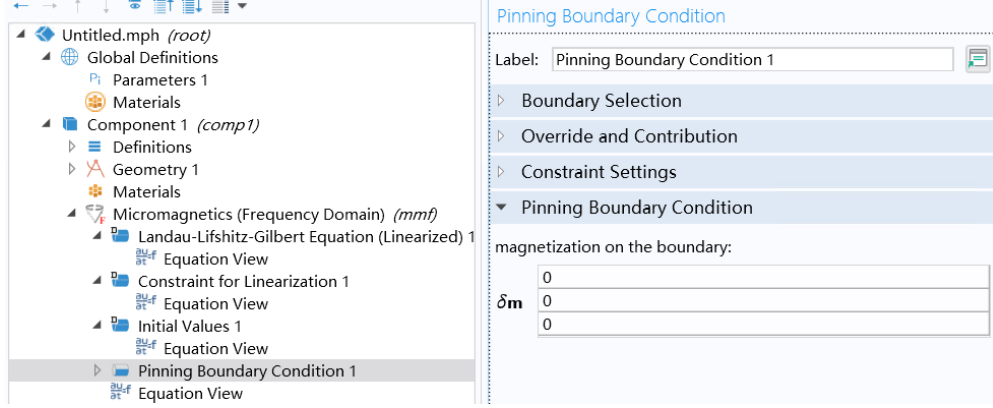


E. Optional Features

We provide two frequently-used boundary conditions for the *Micromagnetics Module (Frequency Domain)*.

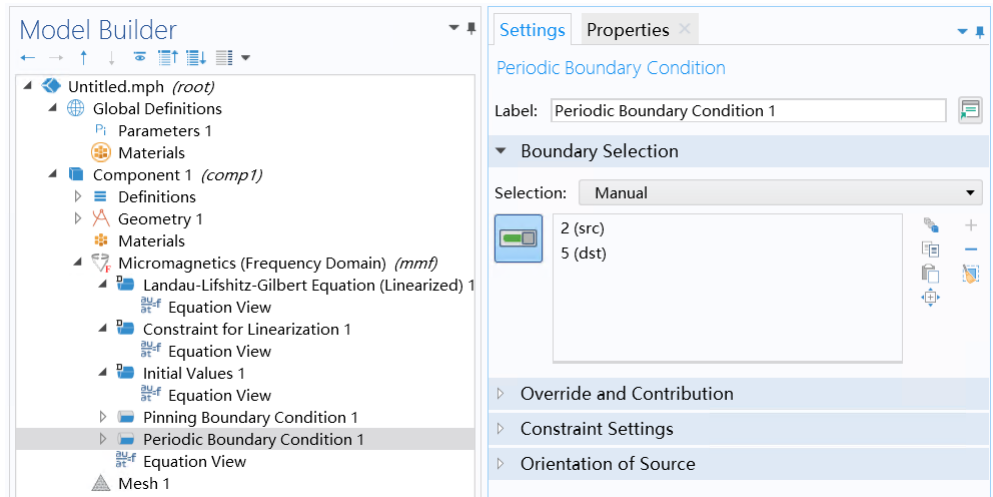
1. Pinning Boundary Condition

By applying **Pinning Boundary Condition**, the magnitude of dynamical excitation can be fixed. The default values are $\delta\mathbf{m} = \{0, 0, 0\}$, corresponding to the scenario that the magnetization is



frozen at the boundary.

2. Periodic Boundary Condition



Periodic Boundary Condition forces the dependent values $\delta \mathbf{m}$ to be continuous on source (src) and destination (dst) boundaries, which is extremely useful when studying periodic structures such as magnonic crystals.

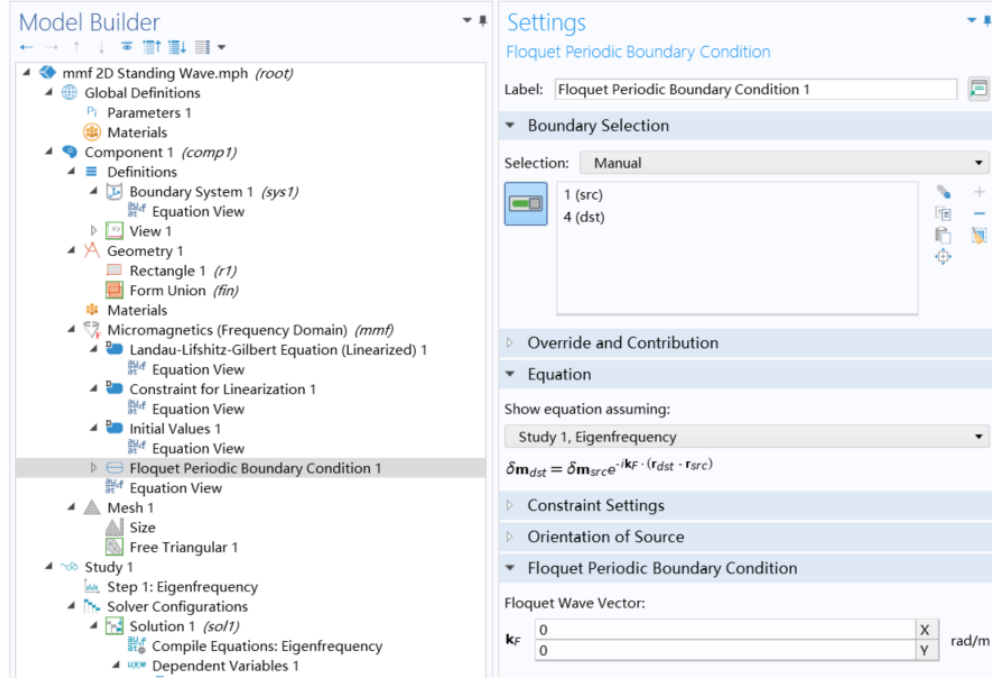
3. Floquet Periodic Boundary Condition

A more general periodic boundary condition is called **Floquet Periodic Boundary Condition**, where one can specify arbitrary phase difference of dependent variables on the source (src) and destination (dst) boundaries with the following form,

$$\delta \mathbf{m}_{\text{dst}} = \delta \mathbf{m}_{\text{src}} e^{-i \mathbf{k}_F \cdot (\mathbf{r}_{\text{dst}} - \mathbf{r}_{\text{src}})}, \quad (20)$$

where $\mathbf{k}_F$ is the **Floquet Wave Vector** (k-vector for Floquet periodicity), $\mathbf{r}_{\text{src}}$ and $\mathbf{r}_{\text{dst}}$ are position of the boundaries. This feature is especially useful for **Eigenfrequency** study, with which one can obtain eigenfrequencies while sweeping wave vector. For example, band structure of magnonic crystals can be efficiently calculated by simulating the minimal element.

The **Floquet Periodic Boundary Condition** works for 1D, 2D and 3D modelling.

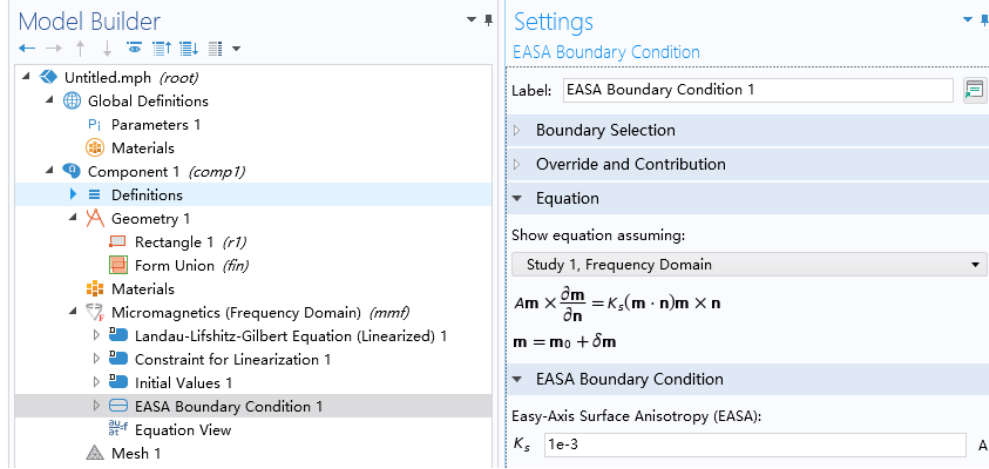


4. Easy-Axis Surface Anisotropy (EASA) Boundary Condition

This boundary condition is a frequency-domain version of the one introduced in Sec. IVE 5, which is included into the module since V2.12. The function of the EASA boundary condition is to simulate materials with certain surface anisotropy, whose easy-axis is along the norm of the surface. The boundary condition is expressed analytically as

$$A \mathbf{m} \times \frac{\partial \mathbf{m}}{\partial \mathbf{n}} = K_s (\mathbf{m} \cdot \mathbf{n}) \mathbf{m} \times \mathbf{n}, \quad (21)$$

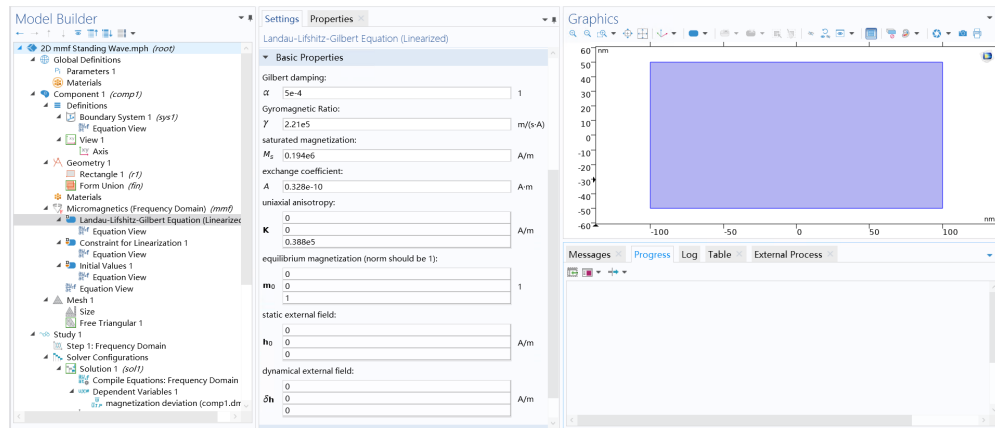
with K_s the EASA strength whose dimension is A (Ampere) and $\mathbf{m} = \mathbf{m}_0 + \delta \mathbf{m}$ according to linear approximation applied in frequency domain. The **EASA Boundary Condition** in frequency domain works for 2D and 3D modelling.



F. Example: Standing Spin Waves in a Magnetic Film

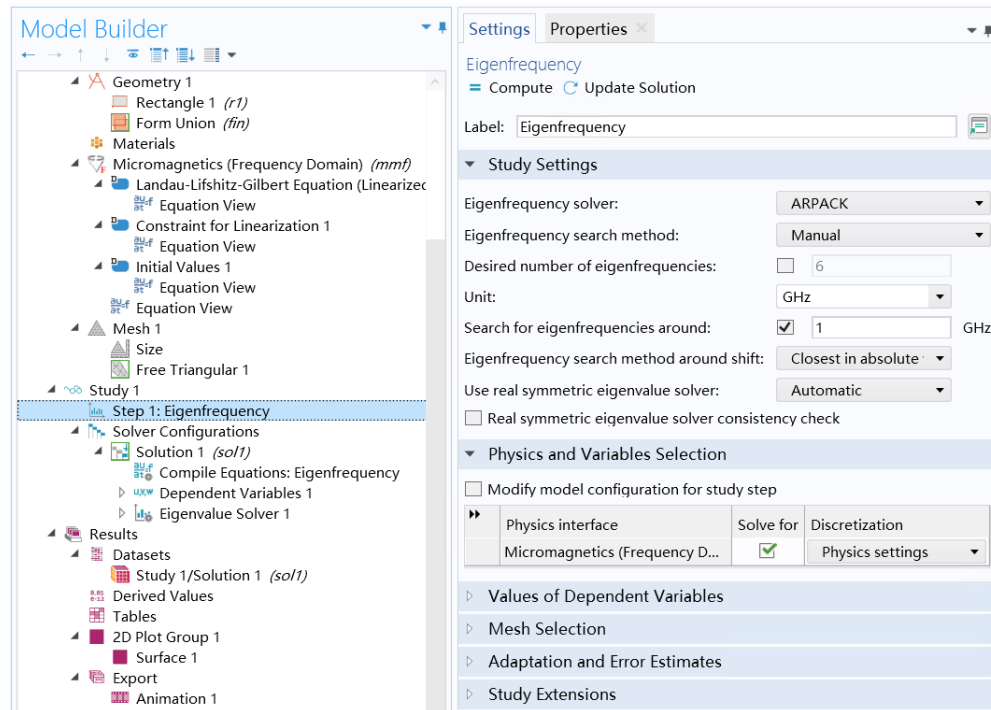
The first example to be presented is the standing spin waves in a 2D magnetic film. The eigenmodes of standing spin waves (eigenfrequency and corresponding profile) can be analyzed in *Micromagnetics Module (Time Domain)* by exciting the magnetic film with a field pulse and performing Fourier transformation on the extracted temporal and spatial data. The eigenstates excited depends on the symmetry of the external pulse, *e.g.*, you can never excite an even mode with a pulse in odd symmetry. Therefore, the time-domain approach is computational inefficient, although it can to the simulate the realistic system to the utmost extent.

Here, we demonstrate the power of the *Micromagnetics Module (Frequency Domain)*, which is naturally suitable for tasks such as searching for eigenstates and sweeping spectrum.

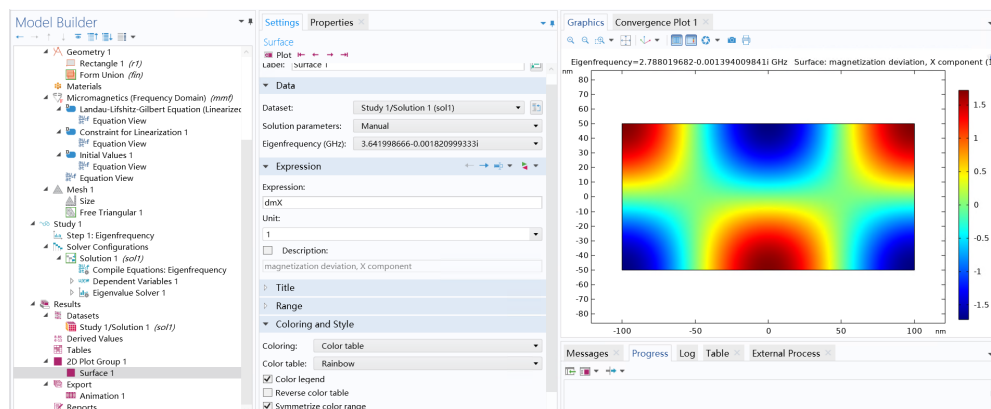


Considering a 2D magnetic film which is magnetized along $+z$ direction, only exchange interaction (to generate spin waves) and uniaxial anisotropy (along z direction, to stabilize the equilibrium magnetization) are activated. Set the **Equilibrium Mangetization** $\mathbf{m}_0$ as $\{0, 0, 1\}$ to make it con-

sistent with the ground state with considered interactions. This is all for the setup in the physics interface.



Next, we move to **Study** and import the **Eigenfrequency** study step. You can change the **Unit** from **Hz** to **GHz** since the dynamics of ferromagnetic materials is usually among GHz regime. You can also check the box of **Search for eigenfrequencies around** and set the desired frequency, which can be estimated from the materials parameters. The other settings are recommended to remain default. Set up an appropriate mesh and click **Compute** to run simulation.



The solution from **Eigenfrequency** study step contains eigenfrequencies (complex number) and its corresponding eigenvectors (profile for spin wave excitation). The real part of eigenfrequency

indicates the resonance point, while its imaginary part is related to the linewidth of the mode mainly induced by Gilbert damping.

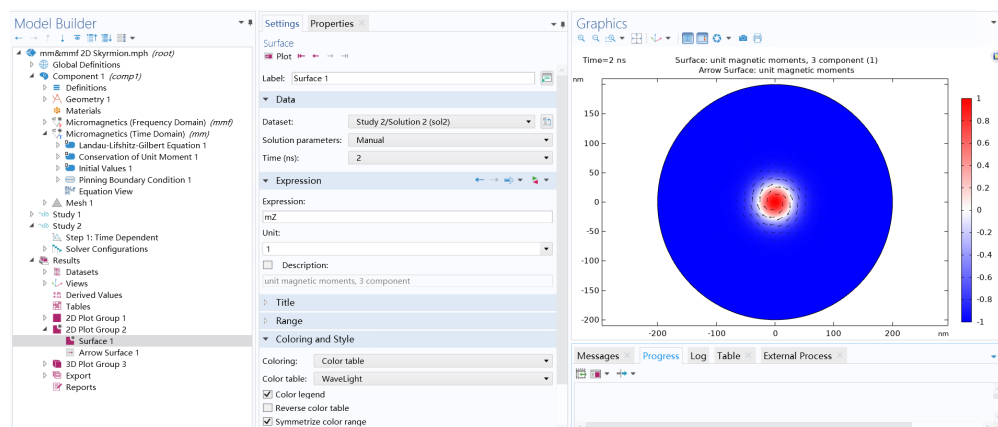
One can also perform sweeping of frequency using **Frequency Domain** step, where you one can define the frequencies of perturbation and obtain response of the system. Spectrum can be plotted by sweeping different frequencies (horizontal axis) and estimating certain global quantities (vertical axis) such as integral of magnetic excitation over whole sample.

Tutorial video for this example can be found [here](#).

G. Example: Eigenmodes of a Magnetic Skyrmion

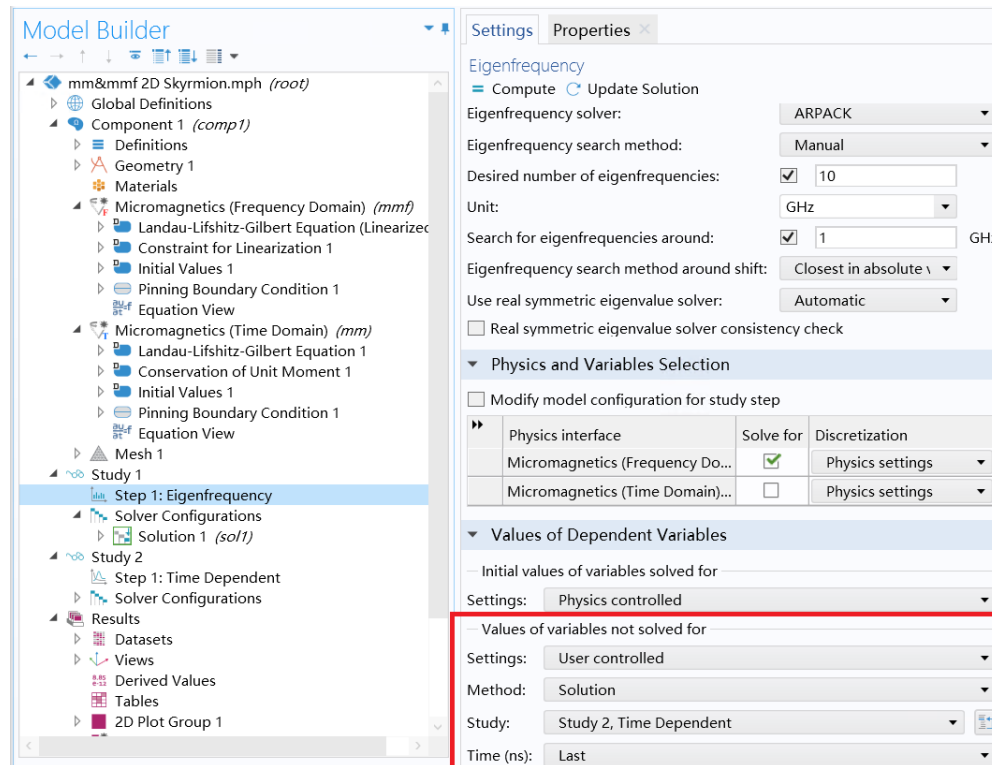
The second example demonstrates the connectivity between *Micromagnetics Module (Time Domain)* and *Micromagnetics Module (Frequency Domain)*. In general, the profile of magnetic textures cannot be expressed by explicit expressions, but can be obtained numerically via micromagnetic simulation. By inheriting solutions from *Micromagnetics Module (Time Domain)*, we are able to perform **Eigenfrequency** and **Frequency Domain** study based on arbitrary complex textures.

Step 1. Considering a Bloch-type 2D magnetic skyrmion, the equilibrium magnetization profile can be stabilized numerically via *Micromagnetics Module (Time Domain)* (for details please visit online [tutorial video](#)). We assume that the user has run the simulation successfully and the solution is already stored in the project file.



Step 2. In order to inherit the solution from *Micromagnetics Module (Time Domain)* to *Micromagnetics Module (Frequency Domain)*, one needs to modify settings in the **Study**. According to the figure below, find **Eigenfrequency (Frequency Domain)**→**Values of Dependent**

Variables→**values of variables not solved for**. Select the following items in the list: **Settings**→User controlled, **Method**→Solution, **Study**→Study 2, Time Dependent (the study you used for *Micromagnetics Module (Time Domain)*) and **Time(ns)**→Last (the time step you want to inherit).



Step 3. Go to the main node of *Micromagnetics Module (Frequency Domain)* and set three components of **Equilibrium Magnetization** $\mathbf{m}_0$ as $\{mX, mY, mZ\}$ respectively, where $\{mX, mY, mZ\}$ are the dependent variables in the *Micromagnetics Module (Time Domain)*. According to **Step 2**, you have defined that the values of $\{mX, mY, mZ\}$ are solution of the last time step in the time-dependent simulation.

Step 4. Now you have defined the **Equilibrium Magnetization**, which is the profile of a magnetic skyrmion. Following the same procedures, you can investigate the eigenmodes of skyrmion such as breathing mode (not shown) and rotation mode (shown in the figure below). Tutorial video for this example can be found [here](#).

Warning: Make sure that the parameters as well as settings of effective fields are consistent for the time- and frequency-approach. Otherwise the solution may not be physically correct.

Model Builder

- mm&mmf 2D Skyrmion.mph (root)
 - Global Definitions
 - Component 1 (comp1)
 - Definitions
 - Geometry 1
 - Materials
 - Micromagnetics (Frequency Domain) (mmf)
 - Landau-Lifshitz-Gilbert Equation (Linearized) 1**
 - Constraint for Linearization 1
 - Initial Values 1
 - Pinning Boundary Condition 1
 - Equation View
 - Micromagnetics (Time Domain) (mm)
 - Landau-Lifshitz-Gilbert Equation 1
 - Conservation of Unit Moment 1
 - Initial Values 1
 - Pinning Boundary Condition 1
 - Equation View
 - Mesh 1
 - Study 1
 - Step 1: Eigenfrequency
 - Solver Configurations
 - Solution 1 (sol1)
 - Study 2
 - Step 1: Time Dependent
 - Solver Configurations
 - Results
 - Datasets
 - Views
 - Derived Values
 - Tables
 - 2D Plot Group 1

Settings **Properties**

Landau-Lifshitz-Gilbert Equation (Linearized)

Basic Properties

Gilbert damping:

α 5e-4 1

Gyromagnetic Ratio:

γ 2.21e5 m/(s·A)

saturated magnetization:

M_s 0.194e6 A/m

exchange coefficient:

A 0.328e-10 A·m

uniaxial anisotropy:

K 0 0.388e5 A/m

equilibrium magnetization (norm should be 1):

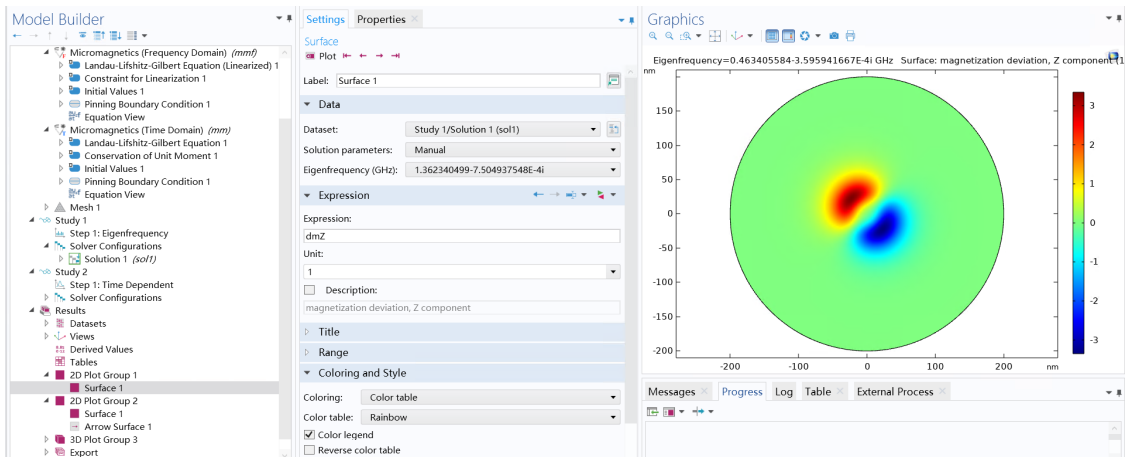
m_0 mX mY mZ 1

static external field:

h_0 0 0 0 A/m

dynamical external field:

δh 0 0 0 A/m



VI. BUILT-IN VARIABLES AND CONSTANTS

Ohter than variables and constants built in COMSOL Multiphysics, there are new ones defined in *Micromagnetics Module*, which are listed below in case users may invoke them when customizing effective fields or torques.

Table I: Shared built-in variables and constants in *Micromagnetics Module*

Name	Symbol	Dimension	Description
alpha	α	1	Gilbert damping
gamma	γ	Hz/(A/m)	gyromagnetic ratio
Ms	M_s	A/m	saturated magnetization
A	A	A·m	exchange coefficient
K	K	A/m	anisotropy strength
eK	$\mathbf{e}_K = \{e_K^x, e_K^y, e_K^z\}$	1	anisotropy axis
P	P	1	spin polarization factor
jX, jY, jZ	$\mathbf{j} = \{j_x, j_y, j_z\}$	A/m ²	charge current density
beta	β	1	non-adiabatic coefficient
muB	μ_B	A·m ²	Bohr magneton (hidden in interface)
D	D	A	DMI coefficient
Dbtype	-	Boolean	bulk-type DMI (checkbox)
Ditype	-	Boolean	interfacial-type DMI (checkbox)
ActSTT	-	Boolean	activating STT
Ks	K_s	A	Easy-Axis Surface Anisotropy (EASA)

All effective fields are defined in the dimension of A/m, so the coefficients are defined according to this convention. Take the exchange coefficient A as example, whose conventional dimension is J/m and current dimension is A·m. The transformation can be described as $A_{[A \cdot m]}^{\text{new}} = \frac{2A_{[J/m]}^{\text{old}}}{\mu_0 M_s}$. The same rule apply for other parameters such as uniaxial anisotropy K and DMI coefficient D .

The default prefix for *Micromagnetics Module (Time Domain/Frequency Domain)* is mm/mm.f. When the software detects undefined variables, it's better to try the full name such as mm.mX and mm.f.dmX. The variables can be investigated in **Equation View** (You need to first enable it

Table II: Built-in variables in *Micromagnetics Module (Time Domain)*

Name	Symbol	Dimension	Description
mX, mY, mZ	$\mathbf{m} = \{m_x, m_y, m_z\}$	1	unit magnetization (dependent variables)
H1, H2, H3	$\mathbf{H} = \{H_x, H_y, H_z\}$	A/m	external field (effective field)
tau1, tau2, tau3	$\boldsymbol{\tau} = \{\tau_x, \tau_y, \tau_z\}$	1/s	torques
m_initX, m_initY, m_initZ	-	1	initial value of dependent variables
seed	-	1	random seed
T	T	K	Temperature
ActTem	-	Boolean	activating finite temperature
dV	ΔV	m ³	unit cell volumn
dt	Δt	s	time step taken by the solver
V	V	m ³	volume (only available in 0D model)
I	I	A	electric current (only available in 0D model)

Table III: Built-in variables in *Micromagnetics Module (Frequency Domain)*

Name	Symbol	Dimension	Description
dmX, dmY, dmZ	$\delta\mathbf{m} = \{\delta m_x, \delta m_y, \delta m_z\}$	1	excitation (dependent variables)
m01, m02, m03	$\mathbf{m}_0 = \{m_x^0, m_y^0, m_z^0\}$	1	equilibrium magnetization
h01, h02, h03	$\mathbf{h}_0 = \{h_x^0, h_y^0, h_z^0\}$	A/m	external field (static)
dh1, dh2, dh3	$\delta\mathbf{h} = \{\delta h_x, \delta h_y, \delta h_z\}$	A/m	external field (dynamical)
kFX, kFY, kFZ	$\mathbf{k}_F = \{k_F^x, k_F^y, k_F^z\}$	rad/m	Floquet periodic vector

as shown in the figure below), where you can also modify the weak form of the equation (not recommended).

The screenshot displays the COMSOL Multiphysics Model Builder interface. The left pane shows the model tree with 'Micromagnetics (Time Domain) (mm)' selected. The middle pane shows the 'Settings' for the 'Equation View' of the 'Landau-Lifshitz-Gilbert Equation 1'. The right pane shows the 'Graphics' settings, with the 'Show More Options' button highlighted by a red circle and a red arrow pointing to the 'Equation Sections' option in the expanded menu.

Model Builder

- Untitled.mph (root)
 - Global Definitions
 - Parameters 1
 - Materials
 - Component 1 (comp1)
 - Definitions
 - Geometry 1
 - Materials
 - Micromagnetics (Frequency Domain) (mmf)
 - Landau-Lifshitz-Gilbert Equation (Linearized) 1
 - Constraint for Linearization 1
 - Initial Values 1
 - Equation View
 - Micromagnetics (Time Domain) (mm)
 - Landau-Lifshitz-Gilbert Equation 1
 - Equation View
 - Conservation of Unit Moment 1
 - Initial Values 1
 - Equation View
 - Mesh 1
 - Study 1
 - Step 1: Frequency Domain
 - Solver Configurations
 - Job Configurations
 - Results

Settings

Equation View

Variables

Name	Expression	Unit
mm.alpha	5.0E-4	1
mm.gam...	221000	m/(s·A)
mm.Ms	194000	A/m
mm.A	3.28E-11	A·m
mm.H1	0	A/m
mm.H2	0	A/m

Shape Functions

Name	Shape function	Unit	Description
mX	Lagrange (Qu...	1	unit magnetic moments,
mY	Lagrange (Qu...	1	unit magnetic moments,
mZ	Lagrange (Qu...	1	unit magnetic moments,

Weak Expressions

Weak expression
$d(mX,t)*test(mX)+mm.gamma*(mm.H3*mY-mm.H2*mZ)*$
$d(mY,t)*test(mY)+mm.gamma*(-mm.H3*mX+mm.H1*mZ)$
$d(mZ,t)*test(mZ)+mm.gamma*(mm.H2*mX-mm.H1*mY)*$

Graphics

Show More Options

- Physics
 - ☒ Equation Sections
 - ☒ Equation View
 - ☒ Stabilization
 - ☒ Equation-Based Contributions
 - ☒ Extra Dimensions
 - ☒ Advanced Physics Options
- Study
 - ☒ Batch
 - ☒ Sensitivity
 - ☒ Multigrid Level
 - ☒ Reduced-Order Modeling
 - ☒ Solver and Job Configurations
- Results
 - ☒ Views
- General
 - ☒ Override and Contribution
 - ☒ Variable Utilities
 - ☒ Material Library Settings

VII. MULTIPHYSICS COUPLING

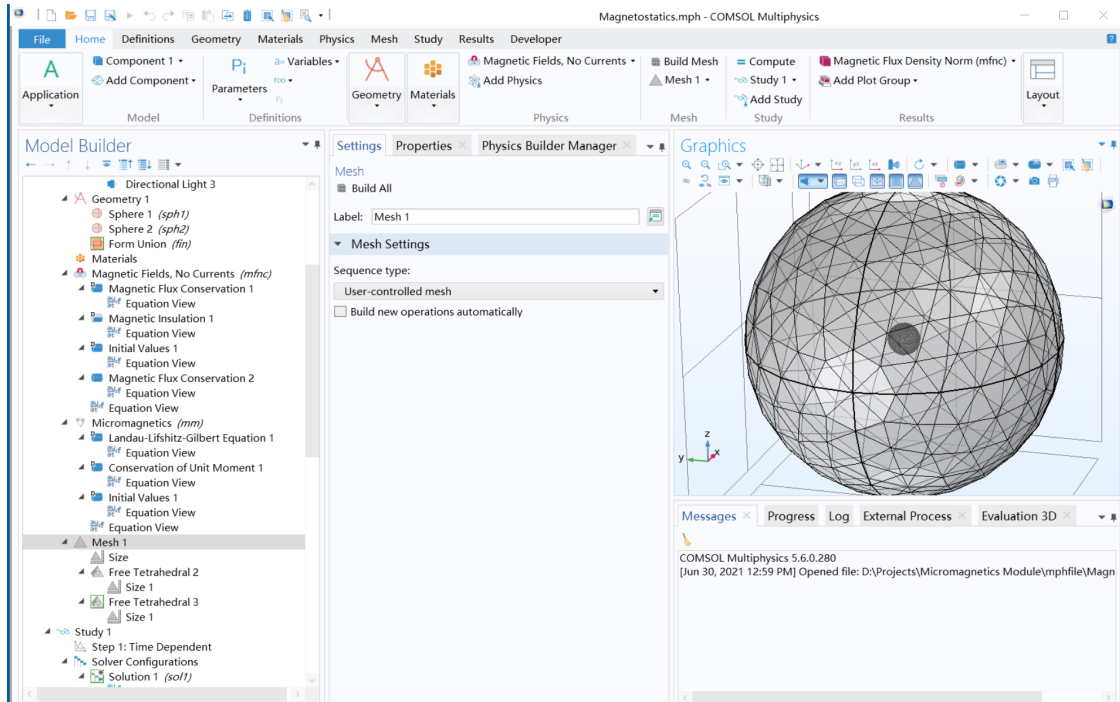
One of the advantages to use COMSOL for micromagnetic simulation is its ability to realize multiphysics coupling. In this section, we demonstrate the possible multiphysics simulation conducted by the *Micromagnetics Module* and other built-in modules.

A. Magnetostatic Coupling (AC/DC Module)

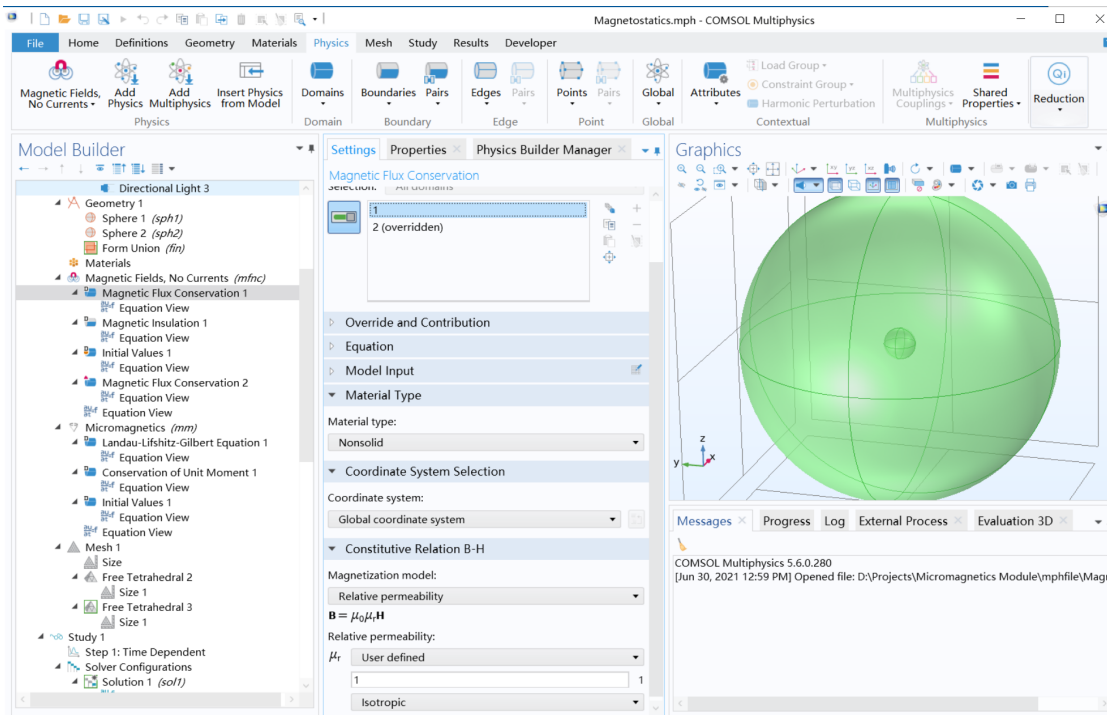
1. Example: Demagnetizing Field in Magnetic Sphere (Coupling in Time Domain)

A magnetic moment is equivalent to a magnetic dipole, which generates dipolar field to the whole space. At the same time, the magnetic moment feels the dipolar fields generated by other moments, resulting in dipole-dipole interaction. Dipolar spin waves can be simulated in the presence of dipole-dipole interaction. Except the *Micromagnetics Module*, another physics interface governing the Maxwell's equations is needed. To simulate magnetostatic problems, the *Magnetic Fields, No Currents (mfnc)* interface in *AC/DC Module* is sufficient.

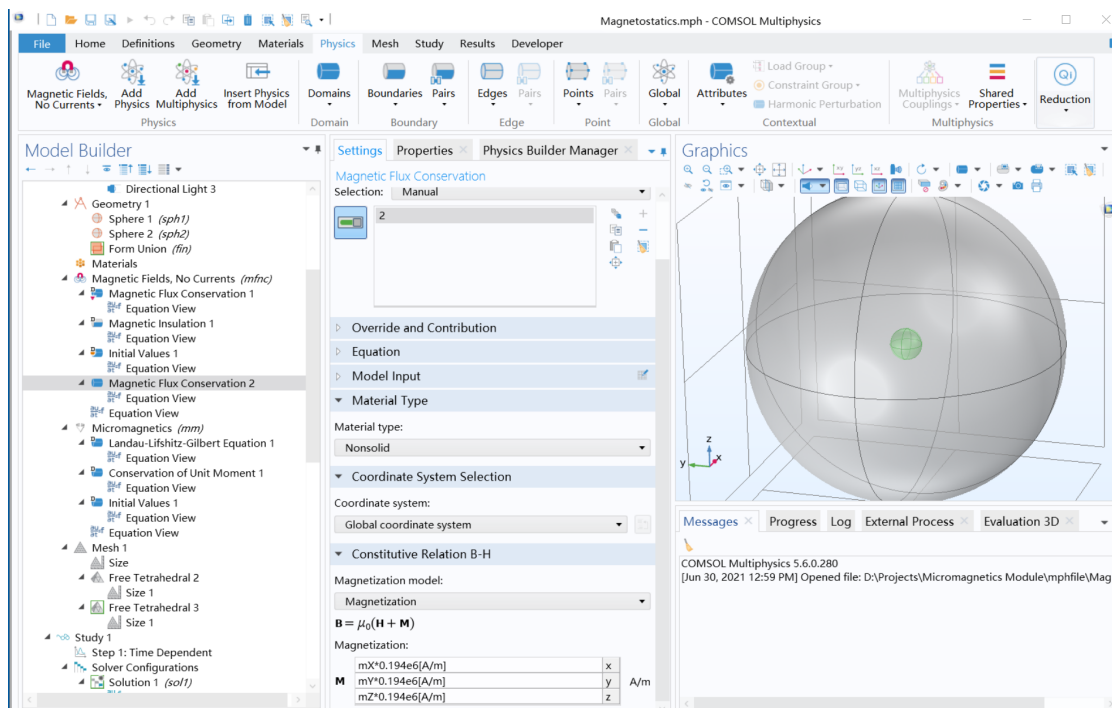
A simple demonstration for magnetostatic coupling is to simulate the demagnetizing field of a magnetized sphere. Add *Magnetic Fields, No Currents (mfnc)* interface and establish the appropriate geometry (two spheres, one for the magnetic sphere and the other one for the air region).



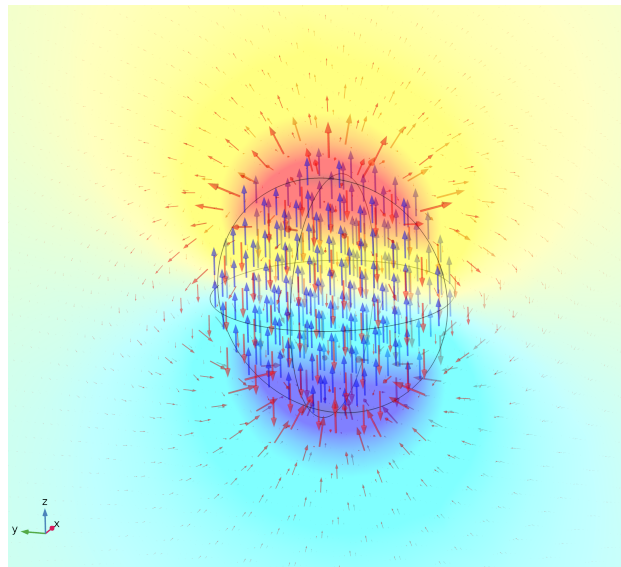
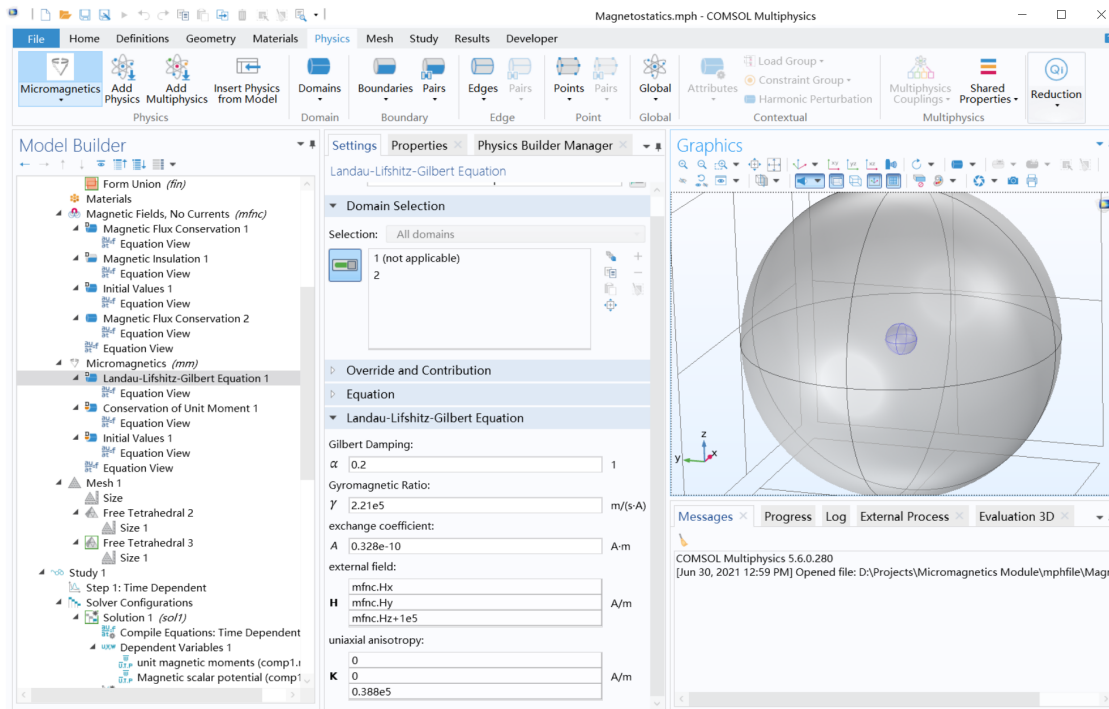
For the **Domain 1** (the air), set the relative permeability μ_r as 1.



For the **Domain 2** (the magnetic sphere), set the magnetization which is the solution from *Micromagnetics Module*, where $M_s = 0.194 \times 10^6$ A/m is the saturated magnetization for YIG. It should be noticed that the dependent variables (m_x, m_y, m_z) are dimensionless.



It's also necessary to import the solutions from *AC/DC Module* to *Micromagnetics Module*. Add terms mfnc.Hx , mfnc.Hy and mfnc.Hz to the **external fields**. An extra static field with magnitude 10^5 A/m is applied in the z direction to stabilize the magnetization inside the sphere. Large **Gilbert damping** is chosen to accelerate the relaxation process.



The simulation shows the self-consistent solutions for magnetization distribution (blue arrows), demagnetizing field (red arrows) and magnetic scalar potential (color). To obtain the solution

successfully, a relaxation process is needed which will be introduced in the next section. Tutorial video for this example can be found [here](#).

Note: If the software reports "Failed to find consistent initial values", try to turn off "Check error estimation" following the steps below: Study 1 (by default) → Solver Configurations → Solution 1 → Time-Dependent Solver 1 → Direct → Error → Check error estimate: **No**.

2. Example: Magnetostatic Modes of Magnetic Thin Film (Coupling in Frequency Domain)

One example is to study magnetostatic modes of a thin film in frequency domain. In order to do that, four modules are needed: *Micromagnetics Module (Time Domain)*, *Micromagnetics Module (Frequency Domain)* and **TWO Magnetic Fields, No Currents**. The main idea is to solve for dynamical spin wave excitations and dynamical demagnetizing field consistently in the presence of a static background of equilibrium magnetization and resulting demagnetizing field. For general users, the simulation will be divided into two steps, similar to the procedure of studying eigenmodes of magnetic skyrmion (Section. [V G](#)):

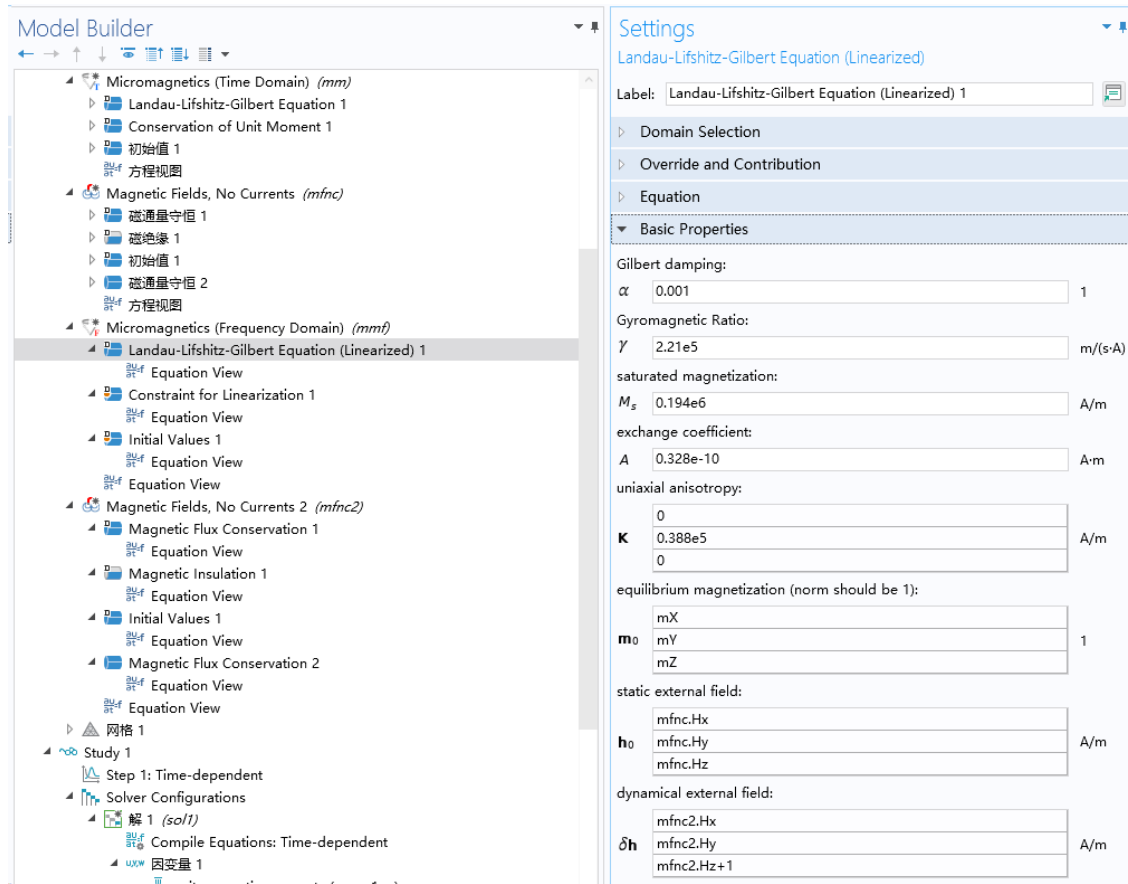
Step 1, Relaxation.

The details are same as discussed in the last section *Coupling in Time Domain* (Section. [VII A 1](#)). The aim is to obtain spatial distribution of static magnetization and static demagnetizing field. The solutions are produced by the *Micromagnetics Module (Time Domain)* and one of the *Magnetic Fields, No Currents*.

Step 2, Simulation in frequency domain.

In the interface of *Micromagnetics Module (Frequency Domain)*, one needs to input parameters exactly same as the one in step 1. Furthermore, one needs to input the solutions from step 1 into **equilibrium magnetization** $\mathbf{m}_0$ ($\{mX, mY, mZ\}$) and **static external field** $\mathbf{h}_0$ ($\{mfnc.Hx, mfnc.Hy, mfnc.Hz\}$). For the **dynamical external field** $\delta\mathbf{h}$, one needs to input the solutions from second *Magnetic Fields, No Currents*, whose default prefix is `mfnc2` and the variables are $\{mfnc2.Hx, mfnc2.Hy, mfnc2.Hz\}$, which are to be solved consistently in the frequency-domain simulation. Note that there is an extra "+1" in the third component of $\delta\mathbf{h}$, which is intended to apply a global excitation with the amplitude 1 A/m and angular frequency ω .

In the interface of second *Magnetic Fields, No Currents*, one needs to define the **Magnetization** as a dynamical one, which are the solutions from *Micromagnetics Module (Frequency Domain)*,



with default name $\{dmX, dmY, dmZ\}$. Before running simulation, remember to confirm the checkboxes (not to check the time-domain modules in the frequency-domain simulation) and inherit the solutions of m_0 and h_0 from step 1.

Model Builder

- Micromagnetics (Time Domain) (*mm*)
 - Landau-Lifshitz-Gilbert Equation 1
 - Conservation of Unit Moment 1
 - 初始值 1
 - 方程视图
- Magnetic Fields, No Currents (*mfnc*)
 - 磁通量守恒 1
 - 磁绝缘 1
 - 初始值 1
 - 磁通量守恒 2
 - 方程视图
- Micromagnetics (Frequency Domain) (*mmf*)
 - Landau-Lifshitz-Gilbert Equation (Linearized) 1
 - Equation View
 - Constraint for Linearization 1
 - Equation View
 - Initial Values 1
 - Equation View
- Magnetic Fields, No Currents 2 (*mfnc2*)
 - Magnetic Flux Conservation 1
 - Equation View
 - Magnetic Insulation 1
 - Equation View
 - Initial Values 1
 - Equation View
 - Magnetic Flux Conservation 2
 - Equation View

Settings
Magnetic Flux Conservation

Label: Magnetic Flux Conservation 2

Domain Selection

Override and Contribution

Equation

Model Input

Material Type

Material type: Nonsolid

Coordinate System Selection

Coordinate system: Global coordinate system

Constitutive Relation B-H

Magnetization model: Magnetization

$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$

Magnetization:

M_x	M_y	M_z
M_x	M_y	M_z
M_x	M_y	M_z

A/m

Model Builder

- Equation View
- Equation View
- 网格 1
- Study 1
- Study 2
 - Step 1: Frequency Domain
 - Solver Configurations
 - Solution 2 (*sol2*)
 - Compile Equations: Frequency Domain
 - Dependent Variables 1
 - magnetization deviation (comp1.dm)
 - unit magnetic moments (comp1.m)
 - Magnetic scalar potential (comp1.Vm)
 - Magnetic scalar potential (comp1.Vm2)
 - Stationary Solver 1
 - Direct
 - Advanced
 - Parametric 1
 - Segregated 1
 - Magnetic Fields, No Currents 2
 - Micromagnetics (Frequency Domain)
 - Iterative 1
 - Incomplete LU
 - Multigrid 1
 - Presmoothing
 - SOR
 - Postsmoothing
 - SOR
 - Coarse Solver
 - Direct

Settings
Frequency Domain

Compute Update Solution

Label: Frequency Domain

Study Settings

Results While Solving

Physics and Variables Selection

Modify model configuration for study step

| Physics interface | Solve for | Discretization |
|------------------------------------|-------------------------------------|------------------|
| Micromagnetics (Time Domain)... | ☐ | Physics settings |
| Magnetic Fields, No Currents (...) | ☐ | Physics settings |
| Micromagnetics (Frequency Do... | ☒ | Physics settings |
| Magnetic Fields, No Currents 2... | ☒ | Physics settings |

Values of Dependent Variables

Initial values of variables solved for

Settings: Physics controlled

Values of variables not solved for

Settings: User controlled

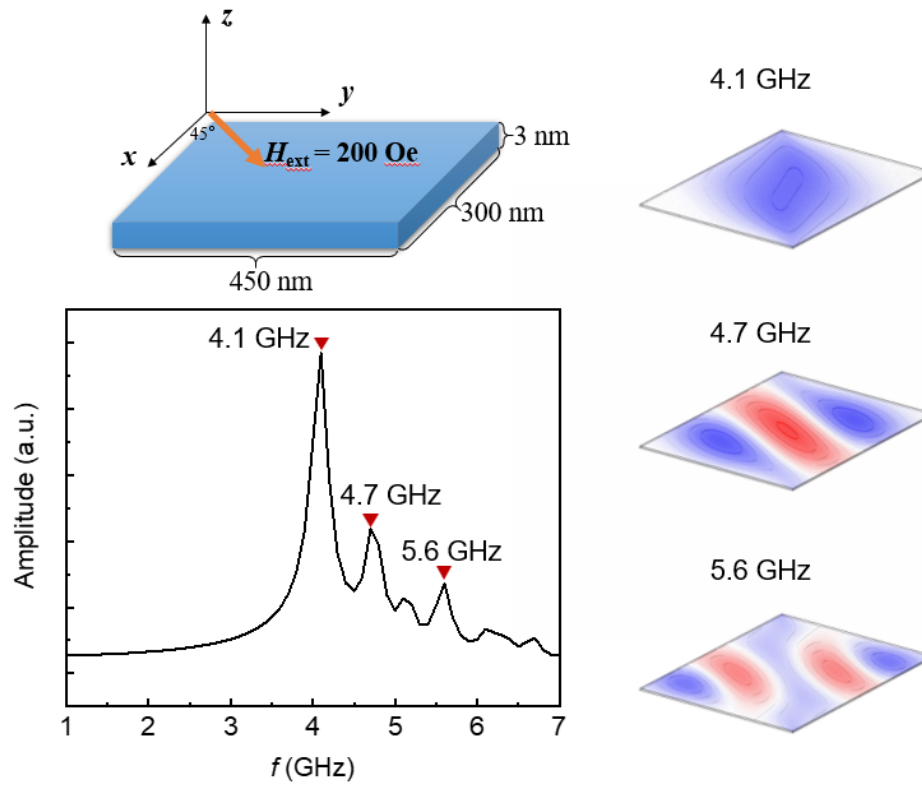
Method: Solution

Study: Study 1, Time-dependent

Time: Last

Store fields in output

Settings: All

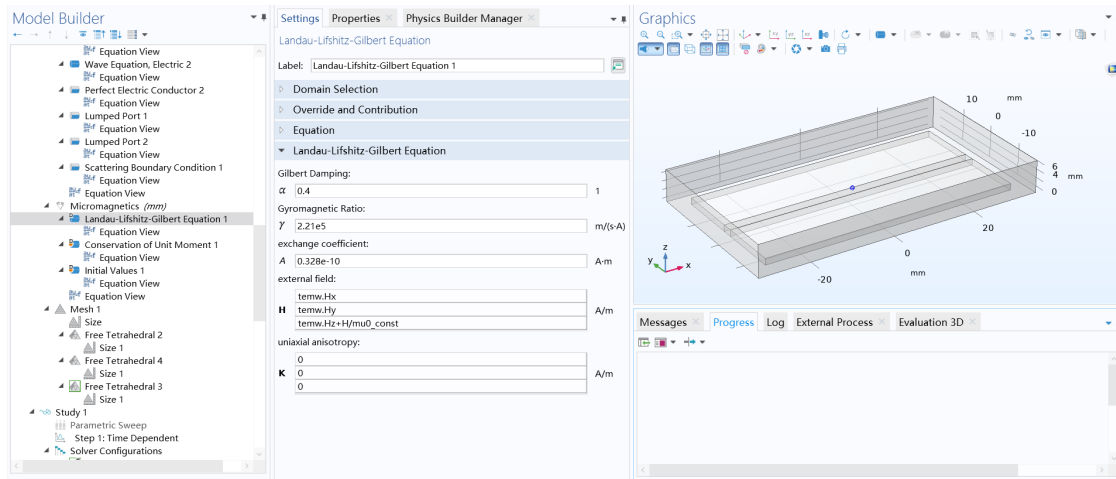


Magnetostatic modes of a magnetic thin film applied with an in-plane field. (Figure and benchmark performed by Dr. Qian Chen from Southeast University)

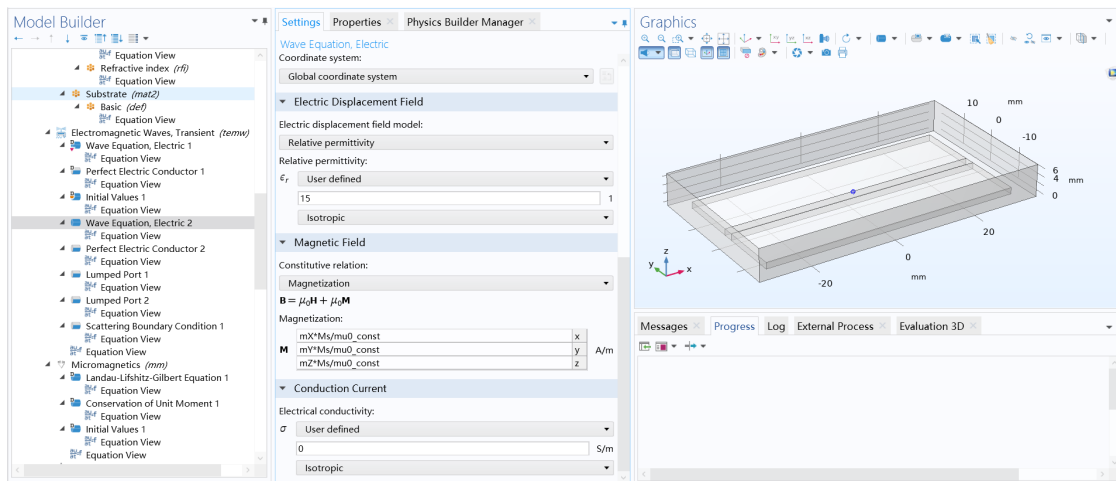
B. Spin Cavitronics (RF Module)

Spin Cavitronics is the interdisciplinary of Spintronics and microwave cavity, where the magnetic excitations are coupled with electromagnetic excitations. In this manner, *AC/DC Module* which applies magnetostatic approximation is no longer sufficient and *RF Module* has to be introduced.

Considering a YIG sphere hanged over a metallic waveguide, input the effective field from *RF Module* into *Micromagnetics Module* with the prefix t_{emw} , so that the dynamics of YIG sphere can be affected by the electromagnetic excitations.



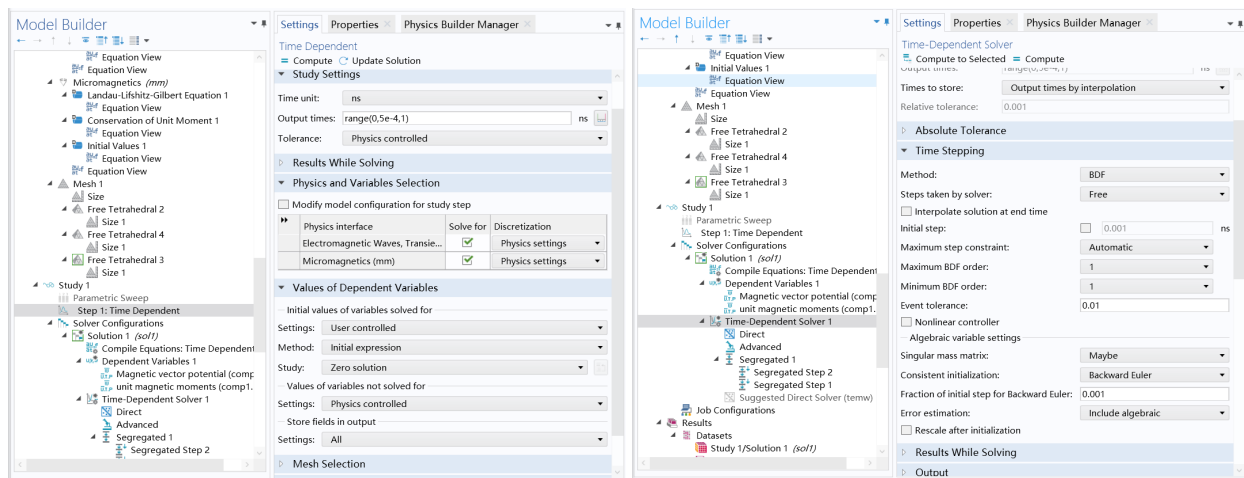
Similarly, input the solution of magnetization from *Micromagnetics Module* into *RF Module*. Remember that a static magnetic moment produces magnetostatic fields, and a dynamic magnetic moment radiates electromagnetic waves.



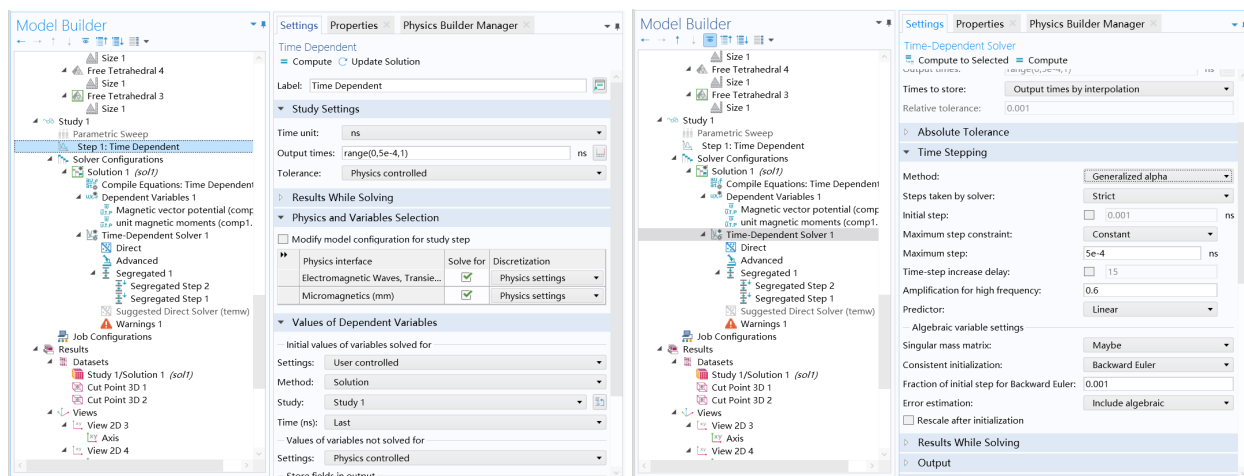
At initial time step, the space is empty (without electromagnetic fields), which is not physical and may induce numerical divergence during initialization. So, there are two steps to run the

time-dependent simulation (same as *AC/DC Module*, although not mentioned in detail in the last section).

Step 1: Relaxation. Take the default expressions as the initial values, *i.e.* magnetization in YIG sphere is aligned in z direction and there is no electromagnetic field distribution in the entire domain. Choose the **Free** method for time stepping, so that the hybrid system can evolve rapidly to the self-consistent solution. In this step, we care about the final state rather than the dynamical process, so we can use a large **Gilbert damping** to accelerate the relaxation process. Here simulation duration is set to 1ns, which is sufficient for a large **Gilbert damping** $\alpha = 0.4$. **Run** the simulation to finish the first step.



Step 2: Excitation. When the first step is finished, we need to inherit the solution at the last time step as the initial values for later simulation. Follow the setups in the figure below and click **Get Initial Value for Step** to inherit the solution.



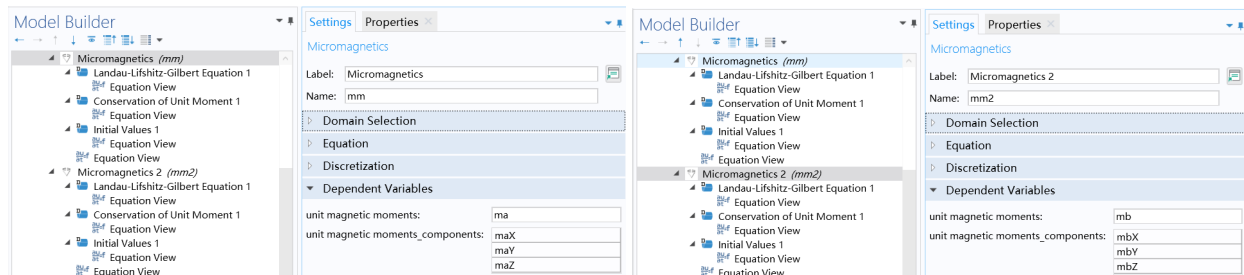
Remember not to clear solutions, otherwise the initial values will also be cleared and you need

to go back to step 1 and "prepare the sample" again. Now you can turn on the **Ports** preset in *RF Module* to inject electromagnetic waves and change the settings for the **time-dependent solver** into *e.g.* **Generalized alpha** with strict steps taken by solver, so that the dynamical process can be simulated in high accuracy. Also remember to reset **Gilbert damping** back to normal values.

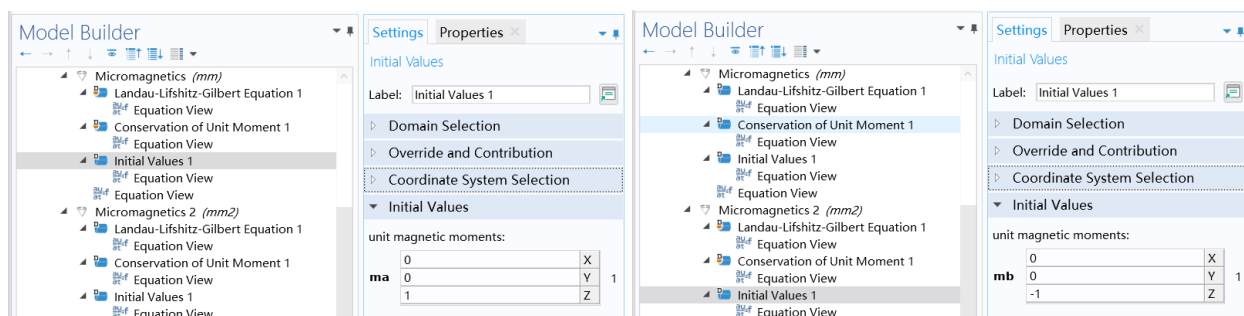
C. Antiferromagnetic Spintronics (Micromagnetics Module $\times 2$)

In antiferromagnetic materials, magnetic moments in neighbour sublattices are aligned in an anti-parallel way. Such a system can be modelled by two coupled LLG equations. *Micromagnetics Module* can couple not only to other physical interfaces, but also to itself. However, since finite elements method is designed for continuum system, such an antiferromagnetic modelling refers to Synthetic Antiferromagnets (SAF).

Import *Micromagnetics Module* two times and you will obtain two sets of independent LLG equations. You can rename the dependent variables such as maX , maY , maZ , mbX , mbY , mbZ respectively for two sub-lattices and system will automatically rename dependent variables in the LLG equation (new feature since V1.32).

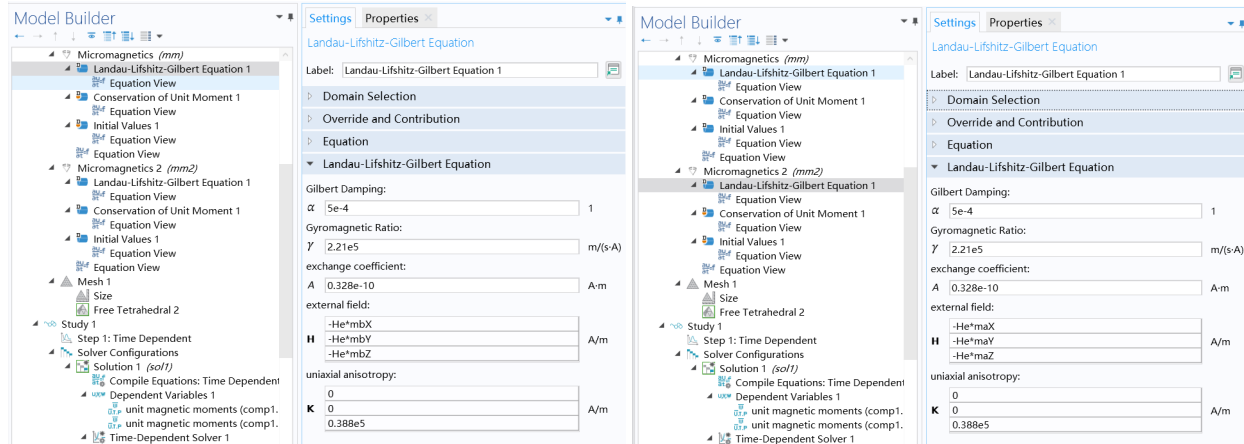


Set the initial state of sub-lattice b to align in $-z$ direction (or *vice versa*).

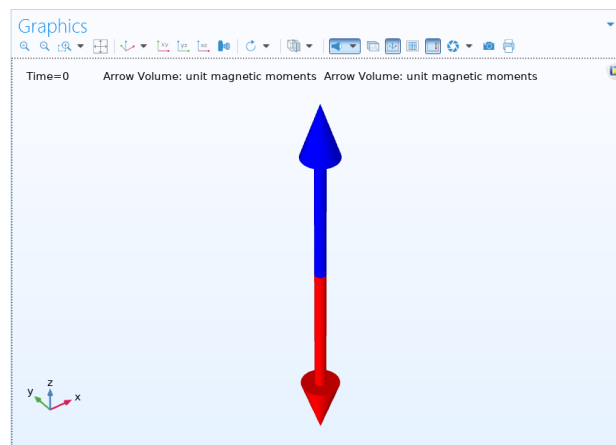


Input extra terms such as $-H_e * mbX$ to introduce inter-layer exchange coupling between two sub-lattices. Here $H_e = 1e5$ [A/m] is defined elsewhere for the strength of coupling and the minus

sign indicates that the coupling is antiferromagnetic, *i.e.* magnetic moments in two sub-lattices tend to align antiparallel. It should be noticed that both $\pm z$ directions are stable in the presence of uniaxial anisotropy.

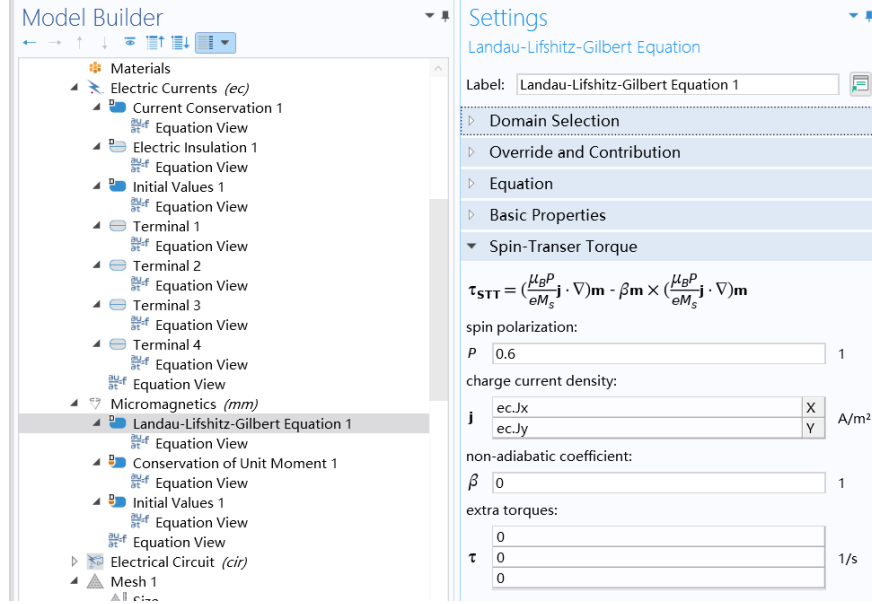


You can visualize the magnetic moments on two sub-lattices by plotting them in arrows with different color. needed which will be introduced in the next section. Tutorial video for this example can be found [here](#).



D. Current-Magnetization Mutual Dynamics (AC/DC Module)

It has been introduced in Section.IV B 3 that current-induced spin-transfer torque can drive magnetization. The charge current density $\mathbf{j}$ can be either input manually or introduced from the solution of *AC/DC Module Electric Currents*. The variables are $ec.J_x$, $ec.J_y$, $ec.J_z$. Such a coupling can be used to simulate texture motion in complex geometries where current density has non-uniform spatial distribution.



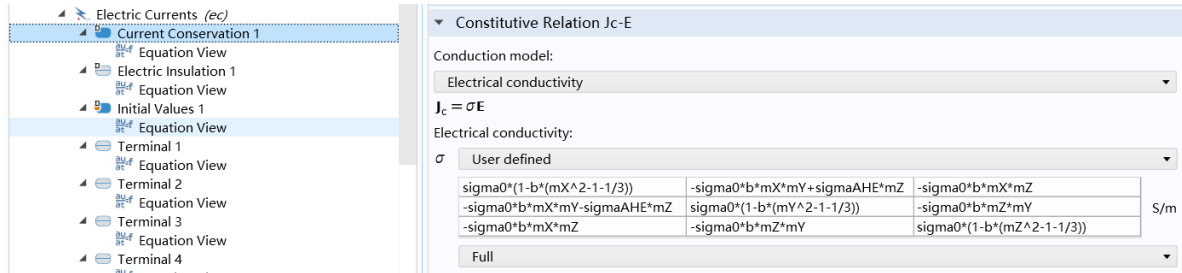
However, such a coupling is unidirectional, which means that the electric currents can affect magnetization, while modified magnetization cannot affect current distribution. In order to realize bidirectional coupling, magnetization-dependent conductivity should be defined, and the electric currents follows the Ohm's law

$$\mathbf{j}(\mathbf{r}) = \hat{\Sigma}[\mathbf{m}(\mathbf{r})] \cdot \mathbf{E}(\mathbf{r}), \quad (22)$$

where $\mathbf{E}$ is electric field and $\Sigma[\mathbf{m}]$ is the magnetization-dependent electrical conductivity:

$$\hat{\Sigma}[\mathbf{m}] = \sigma_{\perp} \mathbb{I} + \sigma_{\delta} \begin{pmatrix} m_x^2 & m_x m_y \\ m_y m_x & m_y^2 \end{pmatrix} + \sigma_{\text{AHE}} \begin{pmatrix} 0 & m_z \\ -m_z & 0 \end{pmatrix}. \quad (23)$$

Here we consider two mechanisms, with $\sigma_{\perp}$ the isotropic conductivity, σ_{δ} induced by Anisotropic Magnetoresistance (AMR), or Planar Hall Effect (PHE), and σ_{AHE} induced by Anomalous Hall Effect (AHE). The definition of a tensor conductivity is supported in the *Electric Currents* interface.



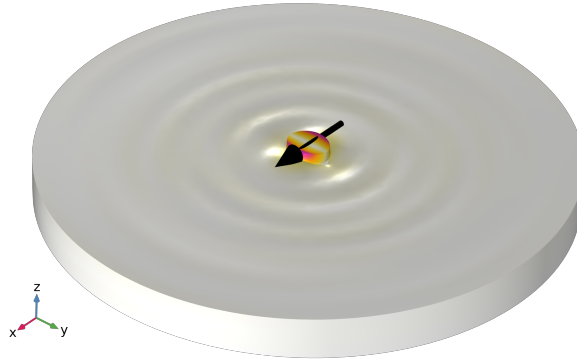
Further details on modelling method can be found in [Physical Review B 104, L180405 \(2021\)](#).

E. Spin Mechanics (Solid Mechanics Module)

In the presence of magneto-elastic coupling (MEC), it's possible for magnetic order to interact with lattice displacement. By combining the *Micromagnetics Module* and the built-in *Solid Mechanics Module*, we can perform multiphysics simulation on Spin Mechanical systems, such as static coupling between magnetic order and elastic deformation (magnetostriction), or dynamical coupling between spin waves and elastic waves (magnon-phonon coupling), facilitating for spin transfer within multiple systems.

1. Example: A Magnetic Disk on Elastic Substrate (Coupling in Time Domain)

One example is shown in the figure below, where a magnetic disk is attached on the surface of a non-magnetic disk. The precession of magnetic moments causes lattice vibration and emits elastic waves. The static magnetostriction is also present, where the magnetic disk is elongated in the direction perpendicular to the magnetic order. The magneto-elastic coupling is bidirectional, which means that magnetic precession can be excited by elastic waves.



Different from magnon-photon coupling, it's not straightforward that magnetic order can be coupled to elastic deformation, so a phenomenological modelling is needed. According to Kittel, *Phys. Rev.* **110**, 836 (1958), the energy density of magneto-elastic coupling for cubic system is

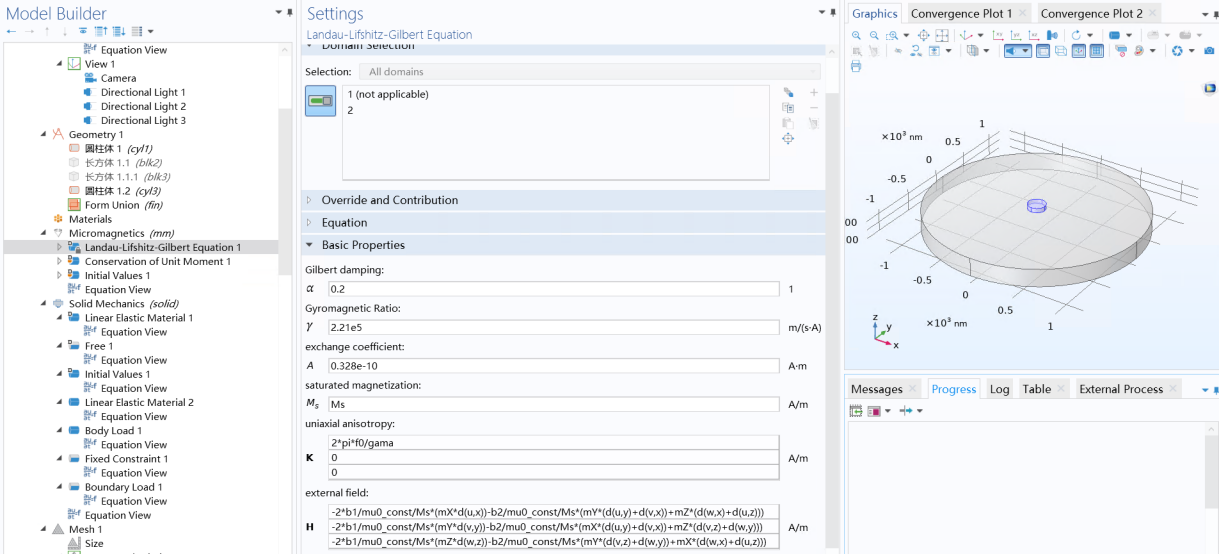
$$\begin{aligned}
 E_{\text{mec}} &= b_1 \sum_i m_i^2 \epsilon_{ii} + b_2 \sum_{i \neq j} m_i m_j \epsilon_{ij} \\
 &= b_1 (m_x^2 \epsilon_{xx} + m_y^2 \epsilon_{yy} + m_z^2 \epsilon_{zz}) + 2b_2 (m_x m_y \epsilon_{xy} + m_y m_z \epsilon_{yz} + m_z m_x \epsilon_{zx}),
 \end{aligned} \tag{24}$$

where b_1 and b_2 are phenomenological coefficients for specific materials and the strain tensor is defined as $\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$ with $\mathbf{u} = \{u_x, u_y, u_z\}$ the lattice displacement away from

its equilibrium position (usually named as u, v, w in COMSOL). According to micromagnetics theory, the effective field induced by magneto-elastic coupling is obtained as

$$\mathbf{H}_{\text{eff}}^{\text{mec}} = -\frac{1}{\mu_0 M_s} \frac{\partial E_{\text{mec}}}{\partial \mathbf{m}} = -\frac{2b_1}{\mu_0 M_s} \begin{pmatrix} m_x \frac{\partial u_x}{\partial x} \\ m_y \frac{\partial u_y}{\partial y} \\ m_z \frac{\partial u_z}{\partial z} \end{pmatrix} - \frac{b_2}{\mu_0 M_s} \begin{pmatrix} m_y \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) + m_z \left(\frac{\partial u_z}{\partial x} + \frac{\partial u_x}{\partial z} \right) \\ m_x \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) + m_z \left(\frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y} \right) \\ m_y \left(\frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y} \right) + m_x \left(\frac{\partial u_z}{\partial x} + \frac{\partial u_x}{\partial z} \right) \end{pmatrix}, \quad (25)$$

which is a three-dimensional vector and can be input into **external field**. This is all for the setup of *Micromagnetics Module*.

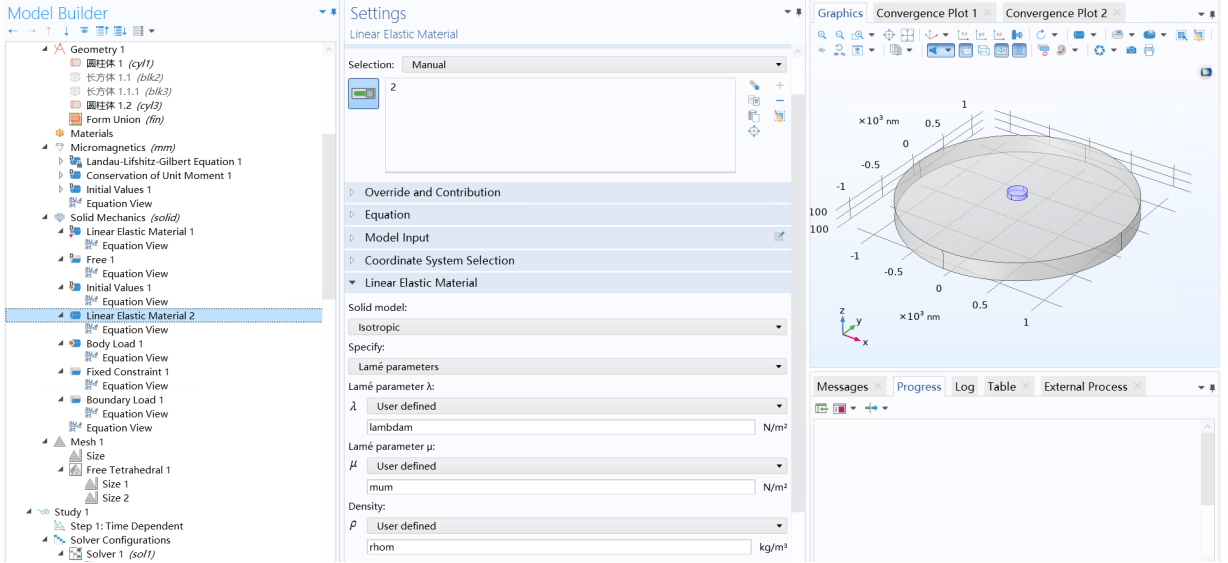
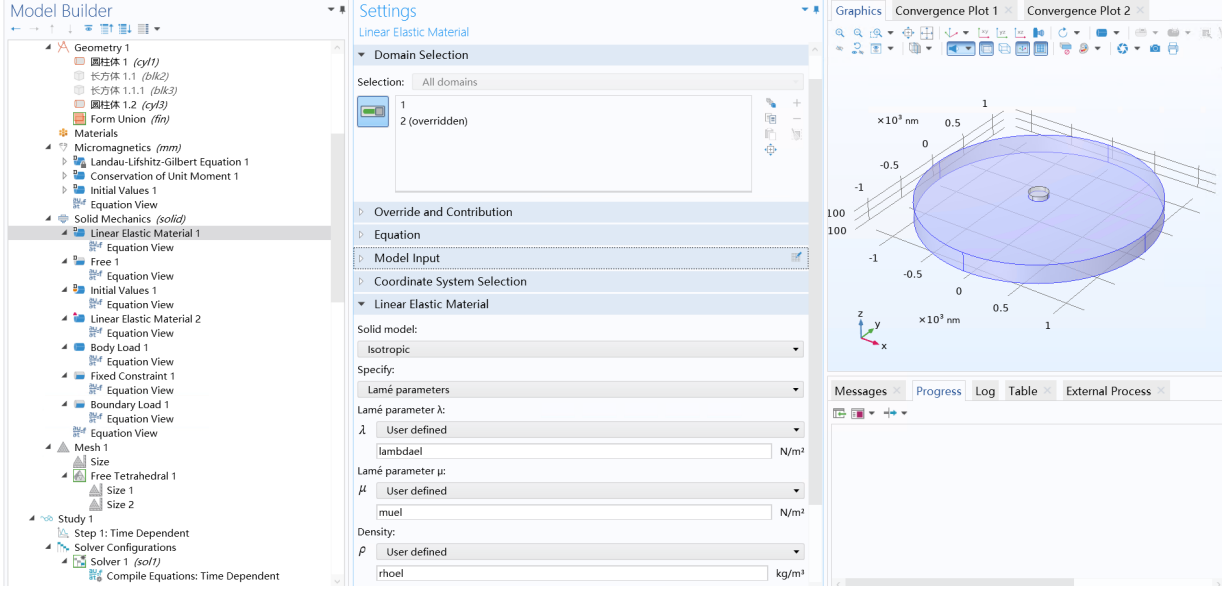


The dynamics of lattice displacement $\mathbf{u}$ is governed by equation of motion for elastic waves

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = \mu \nabla^2 \mathbf{u} + (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) + \mathbf{f}_{\text{mec}} \quad (26)$$

where ρ is mass density and μ, λ are Lamé parameters for specific materials. The contribution from magnetic order comes into the elastic wave equation in the form of effective force $\mathbf{f}_{\text{mec}}$. Users can solve the elastic wave equation via built-in *Solid Mechanics Module*. As shown in the figures below, the material-dependent Lamé parameters should be defined separately within magnetic and non-magnetic materials. You can create two **Linear Elastic Materials** nodes as shown below, or just create one node with the material-dependent parameters ρ, μ, λ .

Similar to the micromagnetics approach, magneto-elastic coupling comes into the elastic wave equation in the form of effective force, which can be obtained by solving Lagrangian equations.

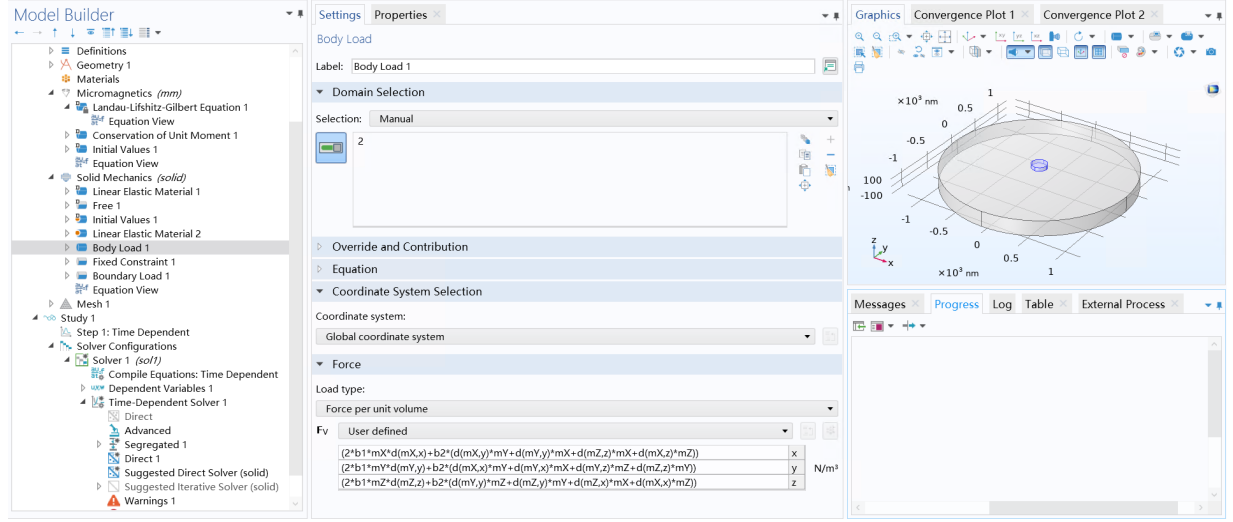


Here we directly show the explicit expression

$$\mathbf{f}_{\text{mec}} = \begin{pmatrix} 2b_1 m_x \frac{\partial m_x}{\partial x} + b_2 \left(\frac{\partial m_x}{\partial y} m_y + \frac{\partial m_y}{\partial y} m_x + \frac{\partial m_z}{\partial z} m_x + \frac{\partial m_x}{\partial z} m_z \right) \\ 2b_1 m_y \frac{\partial m_y}{\partial y} + b_2 \left(\frac{\partial m_x}{\partial x} m_y + \frac{\partial m_y}{\partial x} m_x + \frac{\partial m_y}{\partial z} m_z + \frac{\partial m_z}{\partial z} m_y \right) \\ 2b_1 m_z \frac{\partial m_z}{\partial z} + b_2 \left(\frac{\partial m_y}{\partial y} m_z + \frac{\partial m_z}{\partial y} m_y + \frac{\partial m_z}{\partial x} m_x + \frac{\partial m_x}{\partial x} m_z \right) \end{pmatrix} \quad (\text{in } \Omega). \quad (27)$$

The effective force can be added into the *Solid Mechanics Module* as **Body Load** with the dimension of force per volume, which is active only inside the magnetic materials.

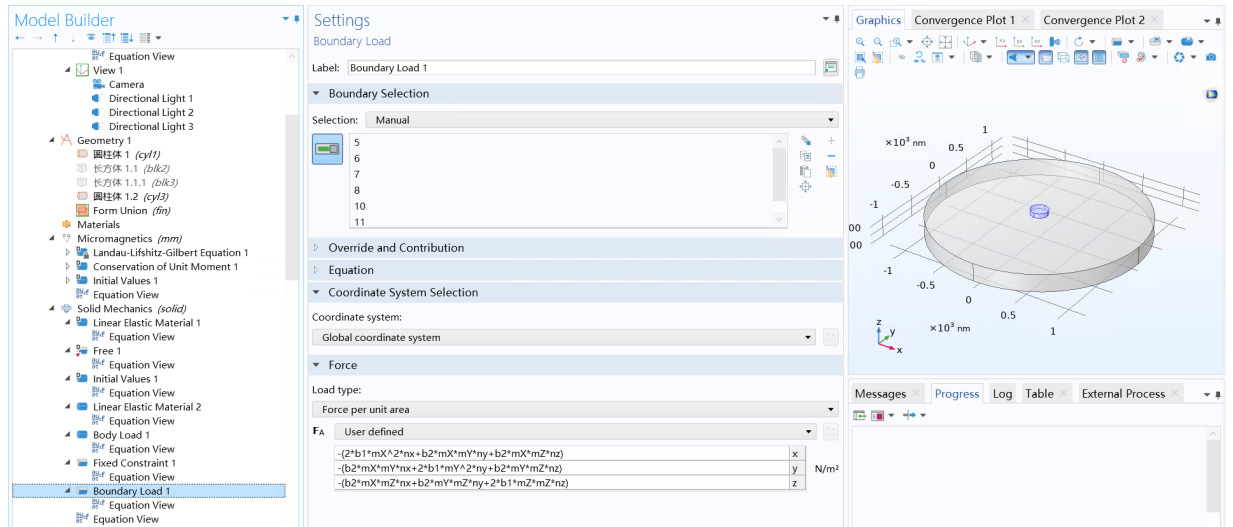
The expression of effective magneto-elastic force indicates that the force is nonzero only when the magnetic order is not uniformly distributed, which is not physical. To simulate the magneto-elastic system correctly, appropriate boundary condition is needed, which can also be obtained by



solving Lagrangian equations,

$$\mathbf{f}_{\text{mec}}^{\text{boundary}} = -\mathbf{n} \cdot \begin{pmatrix} 2b_1m_x^2 & b_2m_xm_y & b_2m_xm_z \\ b_2m_xm_y & 2b_1m_y^2 & b_2m_y m_z \\ b_2m_xm_z & b_2m_y m_z & 2b_1m_z^2 \end{pmatrix} \quad (\text{on } \partial\Omega), \quad (28)$$

where $\mathbf{n} = \{n_x, n_y, n_z\}$ is the norm vector on the boundary. The boundary force can be added into the *Solid Mechanics Module* as **Boundary Load** with the dimension of force per area, which is active only on the boundary of the magnetic material. The boundary condition is general for arbitrary geometry. In order to stabilize the simulation, it's recommended to apply **Fixed Constraint** on the bottom face of the non-magnetic disk.

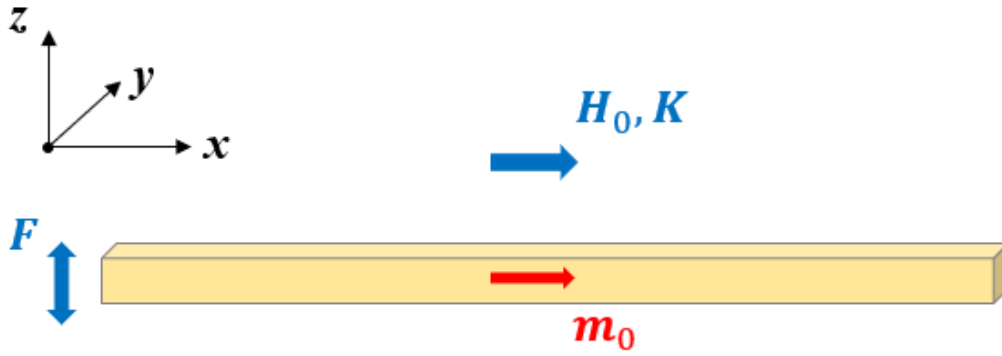


For details, please refer to [J. Phys. Soc. Jpn. 89, 113702 \(2020\)](#) and [Physical Review B 104, 014403 \(2021\)](#). Tutorial video for this example can be found [here](#).

2. Example: Dispersion of Magnon-Polarons (Coupling in Frequency Domain)

Simulation with magneto-elastic coupling can be performed in frequency domain, by combining *Micromagnetics Module (Frequency Domain)* and the built-in *Solid Mechanics Module*.

One example is shown in the figure below, which is a 1D magnetic wire (modelled in 3D with periodic boundary conditions applied). A linear-polarized force is applied on one of the boundaries and drives lattice vibration which further excites the precession of magnetic moments in the presence of magneto-elastic coupling.



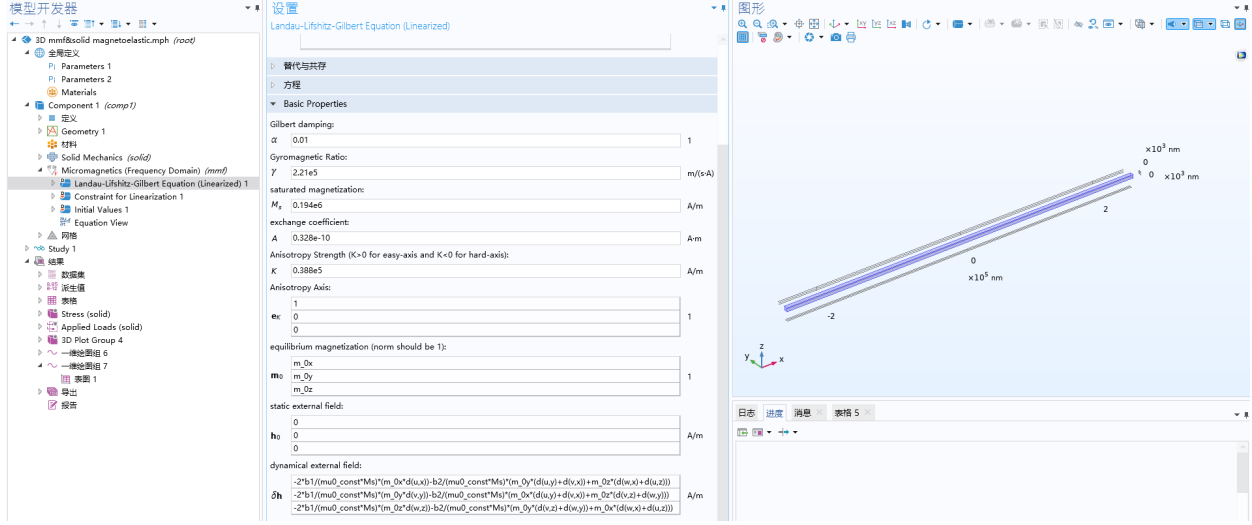
Due to linear approximation $\mathbf{m} = \mathbf{m}_0 + \delta\mathbf{m}e^{i\omega t}$ and similarly for elastic deformation $\mathbf{u} = \mathbf{u}_0 + \delta\mathbf{u}e^{i\omega t}$ in frequency domain, we have $\mathbf{m} \cdot \mathbf{u} = (\mathbf{m}_0 + \delta\mathbf{m}e^{i\omega t}) \cdot (\mathbf{u}_0 + \delta\mathbf{u}e^{i\omega t})$ in the expression of effective field Eq.(25). By neglecting static terms $\mathbf{m}_0 \cdot \mathbf{u}_0$ and high-order terms $\delta\mathbf{m} \cdot \delta\mathbf{u}e^{2i\omega t}$, the effective field induced by magneto-elastic coupling Eq.(25) becomes

$$\mathbf{H}_{\text{eff}}^{\text{mec}} = -\frac{2b_1}{\mu_0 M_s} \begin{pmatrix} m_x^0 \frac{\partial u_x}{\partial x} \\ m_y^0 \frac{\partial u_y}{\partial y} \\ m_z^0 \frac{\partial u_z}{\partial z} \end{pmatrix} - \frac{b_2}{\mu_0 M_s} \begin{pmatrix} m_y^0 \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) + m_z^0 \left(\frac{\partial u_z}{\partial x} + \frac{\partial u_x}{\partial z} \right) \\ m_x^0 \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) + m_z^0 \left(\frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y} \right) \\ m_y^0 \left(\frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y} \right) + m_x^0 \left(\frac{\partial u_z}{\partial x} + \frac{\partial u_x}{\partial z} \right) \end{pmatrix}, \quad (29)$$

which is a three-dimensional vector and can be input into **dynamical external field**. It should be noticed that there is no input for static displacement $\mathbf{u}_0$ in *Solid Mechanics Module (Frequency Domain)*, so the terms $\sim \delta\mathbf{m} \cdot \mathbf{u}_0$ are also eliminated. In general, the static displacement induced by magnetostriction may induce an static effective field to the magnetic system, which can be properly modelled if one customizes the module for elastic waves via *Mathematics Module* or **Physics Builder**.

The equation of motion for elastic waves Eq.(26) can be transformed into frequency domain as

$$-\rho\omega^2 \mathbf{u} = \mu \nabla^2 \mathbf{u} + (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) + \mathbf{f}_{\text{mec}}, \quad (30)$$



with effective force

$$\mathbf{f}_{\text{mec}} = 2b_1 \begin{pmatrix} m_x^0 \frac{\partial \delta m_x}{\partial x} + \delta m_x \frac{\partial m_x^0}{\partial x} \\ m_y^0 \frac{\partial \delta m_y}{\partial y} + \delta m_y \frac{\partial m_y^0}{\partial y} \\ m_z^0 \frac{\partial \delta m_z}{\partial z} + \delta m_z \frac{\partial m_z^0}{\partial z} \end{pmatrix} + b_2 \begin{pmatrix} m_y^0 \frac{\partial \delta m_x}{\partial y} + \delta m_y \frac{\partial m_x^0}{\partial y} + m_x^0 \frac{\partial \delta m_y}{\partial y} + \delta m_x \frac{\partial m_y^0}{\partial y} \\ m_y^0 \frac{\partial \delta m_x}{\partial x} + \delta m_y \frac{\partial m_x^0}{\partial x} + m_x^0 \frac{\partial \delta m_y}{\partial x} + \delta m_x \frac{\partial m_y^0}{\partial x} \\ m_z^0 \frac{\partial \delta m_y}{\partial y} + \delta m_z \frac{\partial m_y^0}{\partial y} + m_y^0 \frac{\partial \delta m_z}{\partial y} + \delta m_y \frac{\partial m_z^0}{\partial y} \end{pmatrix} \quad (31)$$

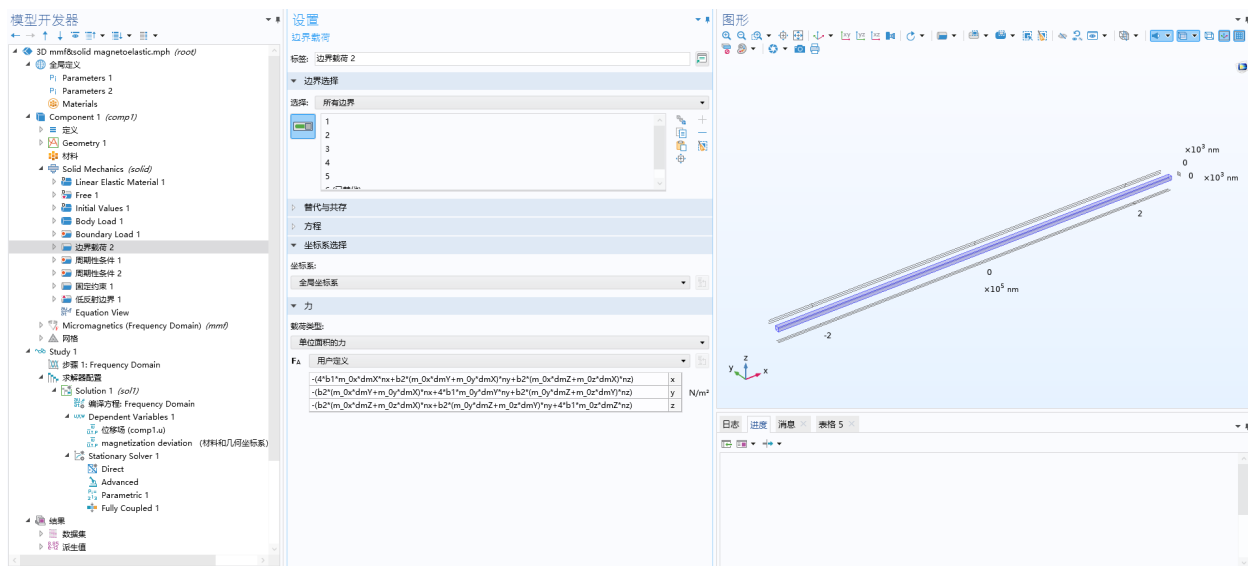
$$+ b_2 \begin{pmatrix} m_x^0 \frac{\partial \delta m_z}{\partial z} + \delta m_x \frac{\partial m_z^0}{\partial z} + m_z^0 \frac{\partial \delta m_x}{\partial z} + \delta m_z \frac{\partial m_x^0}{\partial z} \\ m_z^0 \frac{\partial \delta m_y}{\partial z} + \delta m_z \frac{\partial m_y^0}{\partial z} + m_y^0 \frac{\partial \delta m_z}{\partial z} + \delta m_y \frac{\partial m_z^0}{\partial z} \\ m_x^0 \frac{\partial \delta m_z}{\partial x} + \delta m_x \frac{\partial m_z^0}{\partial x} + m_z^0 \frac{\partial \delta m_x}{\partial x} + \delta m_z \frac{\partial m_x^0}{\partial x} \end{pmatrix}.$$

Static terms $\sim m_0^2$ and high-order terms $\sim \delta m^2$ are neglected in the derivation above. The effective force Eq.(31) can be added into the *Solid Mechanics Module* as **Body Load** with the dimension of force per volume, which is active only inside the magnetic materials.

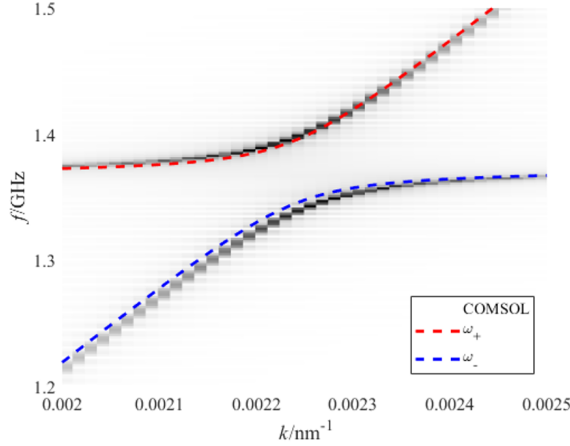
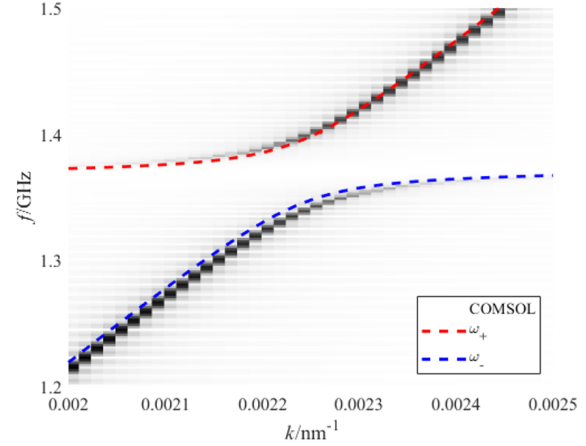
The boundary condition Eq.(28) can be transformed into frequency domain as

$$\mathbf{f}_{\text{mec}}^{\text{boundary}} = -\mathbf{n} \cdot \begin{pmatrix} 4b_1 m_x^0 \delta m_x & b_2(m_x^0 \delta m_y + \delta m_x m_y^0) & b_2(m_x^0 \delta m_z + \delta m_x m_z^0) \\ b_2(m_x^0 \delta m_y + \delta m_x m_y^0) & 4b_1 m_y^0 \delta m_y & b_2(m_y^0 \delta m_z + \delta m_y m_z^0) \\ b_2(m_x^0 \delta m_z + \delta m_x m_z^0) & b_2(m_y^0 \delta m_z + \delta m_y m_z^0) & 4b_1 m_z^0 \delta m_z \end{pmatrix} \quad (32)$$

which can be added into the *Solid Mechanics Module* as **Boundary Load** with the dimension of force per area, and is active only on the interface between magnetic materials and non-magnetic materials.



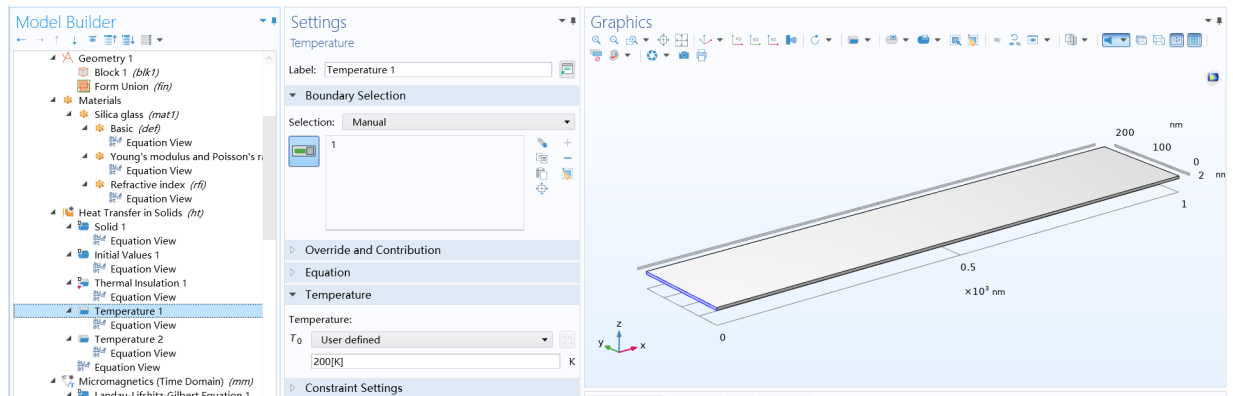
Magneto-elastic waves (MEW), or sometimes called as magnon-polarons around the anticrossing region, can be excited by either force or field, and one can reproduce the dispersion of MEWs and compare with the analytical one (Phys. Rev. B 91, 104409 (2015)). It should be noticed that although static displacement is not considered in *Solid Mechanics Module* in frequency domain, static magnetic components are properly modelled in the *Micromagnetics Module (Frequency Domain)*. So the frequency-domain approach introduced in this section supports studying interaction between excitation of magnetic textures (such as vortex, domain walls and skyrmions) and elastic waves.

(a) The Fourier transformation of dm_Y (b) The Fourier transformation of v

F. Spin Caloritronics (Heat Transfer Module)

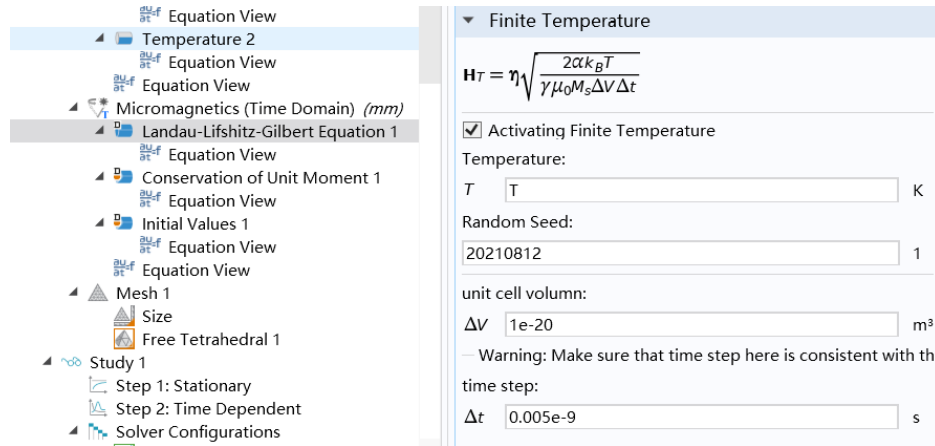
In the presence of finite temperature, the dynamics of magnetic moments (as shown in Section IV B 5) as well as transport of spin currents will be affected by thermal fluctuation. Such an interdisciplinary subject in spintronics called **Spin Caloritronics**. In the following example, we combine *Micromagnetics Module (Time Domain)* and built-in *Heat Transfer in Solids* to demonstrate excitation of spin waves by a thermal gradient.

We start with a 3D model and create an arbitrary slab. The first step is to complete the setup in *Heat Transfer in Solids*. There is nothing special except applying two boundary conditions **Temperature 1** and **Temperature 2** to create a temperature gradient. For the setup in **Solid 1** node, you can input the necessary parameters (heat capacity, density and thermal conductivity) by yourself, or just import them **from material**.

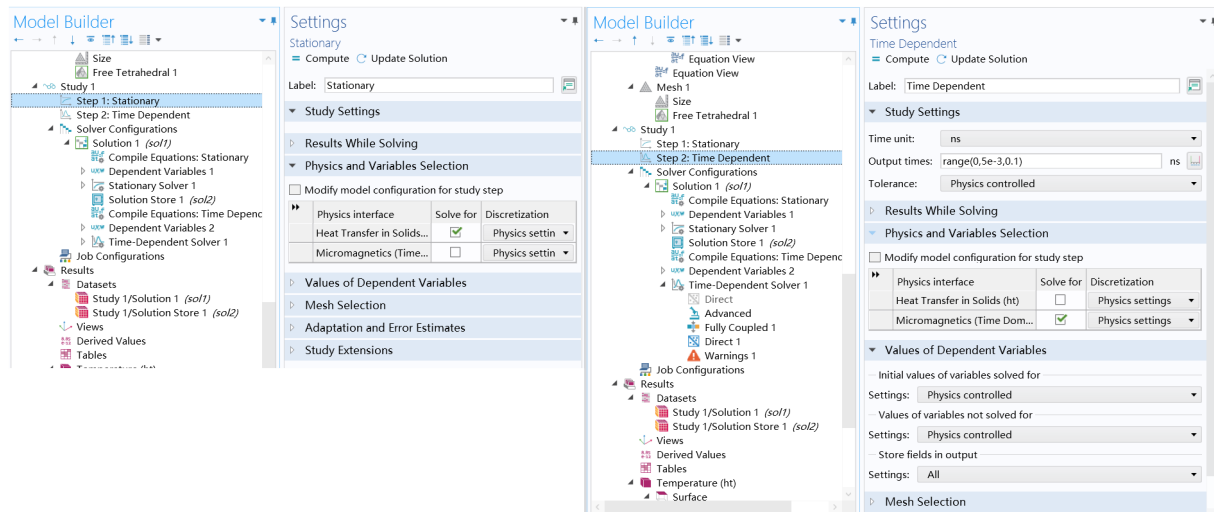


The next step is to complete the setup in *Micromagnetics Module (Time Domain)*. Select the

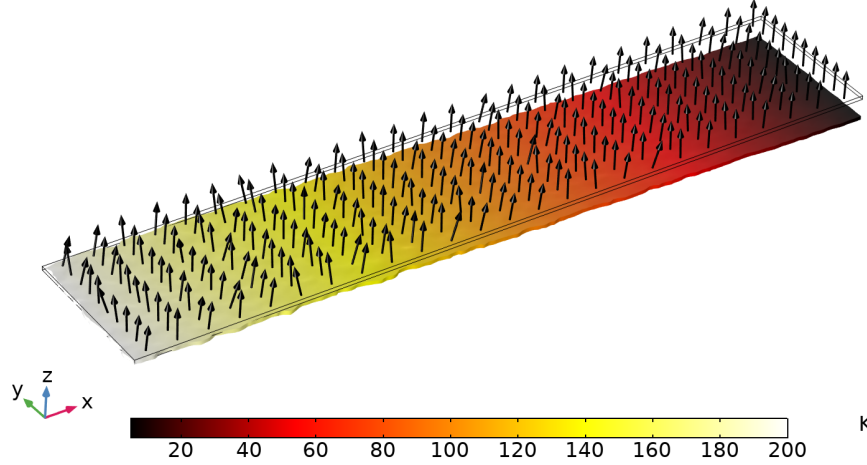
Activating Finite Temperature box, and the only thing you need to do is to input T in the temperature box. Here T is a spatially distributed variable solved by the *Heat transfer Module*. Other settings are same as introduced in Section.IV B 5.



The simulation proceeds with two study steps. The purpose of **Step 1: Stationary** is to solve for spatial distribution of temperature with applied boundary conditions. This step is stationary, but it can be replaced by time-dependent step if the heat transfer process is dynamic. In the next step (**Step 2: Time Dependent**), we solve the LLG equation based on the thermal distribution solved in the last step, and there nothing different with ordinary setup.



With the above settings, we can simulate spin wave excitations driven by thermal gradient as shown below. It's clearly seen that more magnons are excited in the area with higher temperature, hence there is a net magnon flow in opposite to thermal gradient. Since magnons carry spin angular momentum, interesting phenomenon happens such as a **magnetic domain wall** (or a **skyrmion**) move towards the position with higher temperature.



It's important to clarify that the multiphysics coupling discussed here is *unidirectional*, other than *bidirectional* coupling for other interfaces. In the current modelling, thermal distribution can affect the dynamics of the magnetic system, while the state of magnetic orders will not affect thermal distribution reversely. In principle, bidirectional coupling can be modelled since the energy possessed by the precessing magnetic moments will finally be damped into heat via Gilbert damping and it has been observed experimentally that spin waves can emit heat and play as **heat conveyer**. A more thorough modelling will be developed in the future.

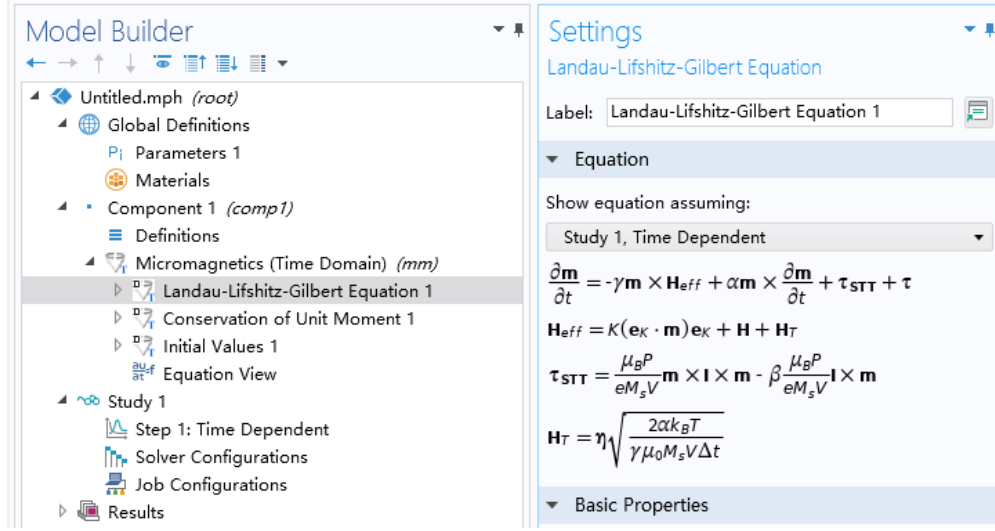
G. MTJ-based Probabilistic Computing (Electrical Circuit Interface in AC/DC Module)

With the help of *Electric Circuit* interface in *AC/DC Module*, users are able to perform SPICE (Simulation Program with Integrated Circuit Emphasis) simulation within COMSOL. According to the concept of multiphysics coupling, one can replace traditional circuit elements by customized devices, such as a magnetic tunnel junction (MTJ), which can be efficiently simulated using the *Micromagnetics Module (Time Domain)* (0D).

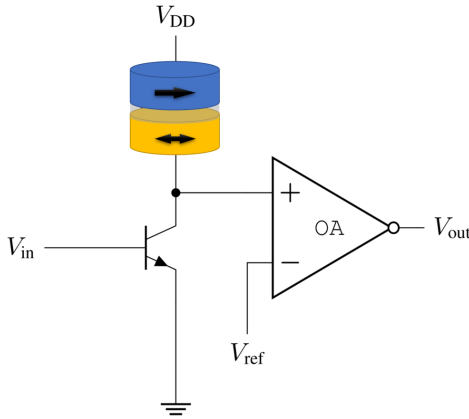
An MTJ is a device with one fixed magnetic layer and one free magnetic layer, which can be treated as a resistor with magnetization-dependent resistance induced by Tunnel MagnetoResistance (TMR) effect. The conductance ($G = 1/R$) of MTJ can be written as

$$G(t) = G_0 \left[1 + \frac{\text{TMR}}{2 + \text{TMR}} m_x(t) \right], \quad (33)$$

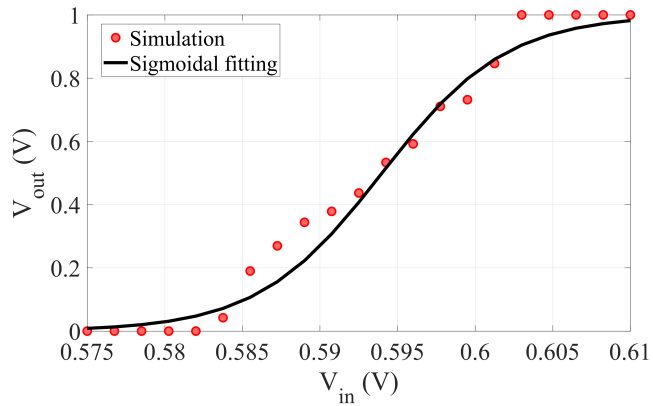
with tunneling magnetoresistance defined as $\text{TMR} = (G_P - G_{AP})/G_{AP}$. Electric current flowing through the MTJ affects the dynamics of magnetization, hence modifying its resistance. The change of resistance further rearrange electric current flowing in the circuits.



The following circuit is composed of an MTJ, an NPN bipolar transistor and a comparator (operational amplifier). Due to the nonlinear characteristics of transistor, the drain voltage V_{drain} drops from V_{DD} to 0 V (ground) gradually with fluctuations. In the presence of a comparator with an inverter, we are able to get binary output whose average value is approximate to *tanh* function or *sigmoid* function.



(a) Circuit diagram of a p-bit



(b) Statistics of V_{out} and sigmoidal fitting

The circuit above is equivalent to a probabilistic bit (p-bit), in analogy to a traditional deterministic bit (d-bit) and a quantum bit (q-bit). By connecting p-bits within a resistive network, one can construct a probabilistic computer, so that advanced computing tasks can be performed such as Bayesian inference, invertible Boolean logic, optimization tasks by classical/quantum annealing. For more information, please check the reference: [p-bits for probabilistic spin logic](#).

VIII. POTENTIAL ISSUES

A. Inhomogeneous Material Properties

As kindly reported by Ms. Jing Xu from Southeast University, there might be an issue when dealing with cases with discontinuous material properties (see the proposed [Standard Problem](#)). As has been tested, when there is a parameter jump, *e.g.*, different exchange stiffness A_i or uniaxial anisotropy K_i corresponding to connecting domains Ω_i , the pinning energy for a moving domain wall will be different compared with conventional micromagnetics simulators such as OOMMF and Mumax (after 3.10). The results produced by the *Micromagnetics Module (Time Domain)* is consistent with the one produced by Mumax 3.9.3. However, the issue is resolved after Mumax 3.10, which is interpreted as an incorrect treatment of saturated magnetization M_s across the boundary ([Mumax 3.10 release notes](#)). However, the saturated magnetization is reduced into other parameters, *e.g.*, $A_{[\text{A}\cdot\text{m}]}^{\text{new}} = \frac{2A_{[\text{J}/\text{m}]}^{\text{old}}}{\mu_0 M_s}$ and we produce correct results for jump of exchange stiffness A . The underlying mechanism is still unclear and this issue remains unsolved, since it might be related to the discretization method embedded in COMSOL.

Therefore, we would like to call awareness for users who are going to perform simulations in the presence of inhomogeneous parameter distribution.

IX. EXAMPLES AND PUBLICATIONS

Here are references and examples accomplished by *Micromagnetics Module*. When you publish your research using this module, you are encouraged to cite the any references below which are helpful to your research.

A. References published by the developers

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“Spin-Wave Diode”

Physical Review X 5, 041049, December, 2015.

†: equal contribution

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[https:](https://cn.comsol.com/paper/3c2a96daab4a69c6bbe7a6a3f91e7968-67443)

[/cn.comsol.com/paper/3c2a96daab4a69c6bbe7a6a3f91e7968-67443](https://cn.comsol.com/paper/3c2a96daab4a69c6bbe7a6a3f91e7968-67443)

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†: equal contribution

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Appointed by J. Phys. Cond. Mat. (JPCM) 2021 Magnonics Roadmap.

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“Spin Cavitronics, multiphysics simulation based on Micromagnetics and RF Module”

[https:](https://cn.comsol.com/paper/93e41ac13c85f542b754fe535eeb86f9-96071)

[/cn.comsol.com/paper/93e41ac13c85f542b754fe535eeb86f9-96071](https://cn.comsol.com/paper/93e41ac13c85f542b754fe535eeb86f9-96071)

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“Unexpected Intermediate State Photoinduced in the Metal-Insulator Transition of Submicrometer Phase Separated Manganites”

Physical Review Letters 120, 267202 (2018), June, 2018

[21] Gongzheng Chen, Jin Lan, Tai Min and Jiang Xiao
 “Narrow waveguide based on ferroelectric domain wall”
Chinese Phys. Lett. 38 087701 (2021)

B. References published by the users

We encourage all users to contact us if your researches are accomplished with the help of this module. We will update the following list timely.

[1] Hanxu Ai and Jin Lan
 “Anatomy of spin-wave-driven magnetic texture motion via magnonic torques”
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Journal of Magnetism and Magnetic Materials, 604, 172315 (2024)

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 “Accurate calculation of magnetic forces on magnetic mineral particles using micromagnetic simulations”
Minerals Engineering 218, 109001 (2024)

X. RELEASE NOTES

V1.0 released in July, 2021

- The first version includes exchange interaction, uniaxial anisotropy and interface for arbitrary effective field and is able to perform multiphysics coupling with *AC/DC Module*, *RF Module* and *Micromagnetics Module* itself.

V1.01

- Interface for arbitrary torque input is added.

V1.1

- A new section **Spin-Transfer Torque** is added and its coupling with *AC/DC Module Electric Currents* is introduced.

V1.2

- DMI with both bulk-type and interfacial-type are added.
- DMI boundary conditions (both bulk-type and interfacial-type) are added.
- Pinning boundary condition is added.
- Periodic boundary condition is added.

V1.3

- Errors in Spin-Transfer Torque section are corrected.
- The checkbox for activating STT is added.
- A new section **Finite Temperature** is added and it's coupling with *Heat Transfer Module* is introduced.

V1.31

- A problem regarding to the easy/hard-axis anisotropy is fixed.

V1.32

- The weak form of LLG equation is modified from scalar representation to a full vector representation, so that the naming problem of dependent variables is fixed when multiple micromagnetics modules are introduced.

V1.33

- Error is fixed on the definition of dependent variables in unit moment conservation constraint.
- A minus sign is added to the bulk-type DMI boundary condition to follow the convention of bulk-type DMI energy function.

V2.0

- The *Micromagnetics Module* was separated to two new sub-modules, *Micromagnetics Module (Time Domain)* and *Micromagnetics Module (Frequency Domain)*.
- The function of *Micromagnetics Module (Time Domain)* is same as the original *Micromagnetics Module* (V1.33).
- A new sub-module, the *Micromagnetics Module (Frequency Domain)* was developed, with which one can perform simulations in frequency domain by solving linearized LLG equation.

V2.01

- For time-dependent simulations using *Micromagnetics Module (Time Domain)*, when coupling to built-in modules such as *Solid Mechanics Module*, an error message appears and states that “Time Derivative Only Allowed With Respect To Mesh Time”. The problem may come from the deformed geometry and meshes to be used by *Solid Mechanics Module*. One solution is to replace the time derivative of dependent variables $\frac{d(m, t)}{dt}$ by $\frac{d(m, TIME)}{dt}$. Now this modification is applied in the current version.

V2.02

- In *Micromagnetics Module (Frequency Domain)*, the convention for the harmonic term $e^{i\omega t}$ is modified to keep consistency with this user’s guide. In early versions such as V2.0 and V2.01, the convention $e^{-i\omega t}$ was used in the module. However, it should be noted that different convention doesn’t lead to different physical results, but the definition of initial phase, as well as the one for chirality of magnetization precession.

- In *Micromagnetics Module (Frequency Domain)*, a typo in **Equation View** was fixed, which doesn't affect the original equation to be solved.

V2.03

- A new type of boundary condition **Easy-Axis Surface Anisotropy (EASA)** is included in the *Micromagnetics Module (Time Domain)* for both 2D and 3D modelling.

V2.04

- A new type of boundary condition **Floquet Periodic Boundary Condition** is included in the *Micromagnetics Module (Frequency Domain)*.

V2.05

- In early versions, thermal fluctuation cannot be activated for 2D and 1D model, because the function `randomnormal(x, y, . . .)` contains unknown variable such as z and y for low-dimensional model. This problem has now been fixed and thermal fluctuation can be activated in 1D, 2D and 3D model.

V2.1

- Modelling in 0D is available in the *Micromagnetics Module (Time Domain)*.

V2.11

- The terms corresponding to uniaxial anisotropy are modified as $K(\mathbf{e}_K \cdot \mathbf{m})\mathbf{e}_K$. In earlier versions, the anisotropy term is $(\mathbf{K} \cdot \mathbf{m})|\mathbf{e}_K|$, which may induce problems when the anisotropy axis is not along pure x, y or z direction.

V2.12

- **EASA Boundary Condition** is available in the *Micromagnetics Module (Frequency Domain)* for both 2D and 3D modelling.

V2.13

- An issue is fixed where variables from different Micromagnetics Modules were being cross-referenced, for example, parameters from the mm2 module were directly calling parameters with the same name in the mm module. This issue did not cause problems in most cases, but when simulating ferrimagnets (where the parameters of two sub-lattices differ), it led to calculation errors.

To be updated

- New formalism such as LLB equation to capture dynamics of magnetization under finite temperature.
- Preset profiles of magnetic textures such as domain walls and skyrmions
- Other types of anisotropy such as cubic anisotropy
- *Ferroelectrics Module* and its coupling with *Micromagnetics Module* in the presence of magneto-electric coupling
- Time-dependent LLG equation in spherical coordinate to accelerate simulation (but may encounter singular points)
- Coupling with Time-dependent Ginzburg-Landau equations to realize bidirectional coupling between magnetic dynamics and superconducting vortexes.

XI. ACKNOWLEDGEMENTS

We thank the following users for their helpful suggestions and interactions.

Ms. HyeongJoo Seo

Daegu Gyeongbuk Institute of Science and Technology

Discussion on implementation of spin-orbit torques

Prof. Junsheng Feng

Hefei Normal University

Reporting typo of equation in COMSOL Blog

Mr. Chengyuan Cai

Huazhong University of Science and Technology

Reporting and testing the issue regarding to coupling between *Micromagnetics Module (Time Domain)* and *Solid Mechanics*

Ms. Jiahui Yuan

Air Force Engineering University

Reporting typo of equations for *Micromagnetics Module (Frequency Domain)*

Dr. Qian Chen

Southeast University

Testing the calculation of magnetostatic eigenmodes of thin film and comparing with OOMMF

Ms. Jing Xu

Southeast University

Reporting and testing the issue regarding to inhomogeneous material properties

Mr. Danrong Xiong

Beihang University

Reporting problem and solution regarding to solver setup in magnetostatic simulations

Micromagnetics Module for COMSOL Multiphysics

Weichao Yu (余伟超)¹, Jiabin Zhang (张家斌)², Chensong Hua (华辰松)¹, and Jiang Xiao (肖江)^{1,2}

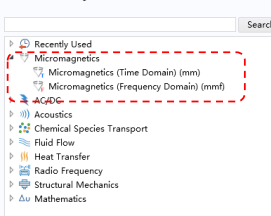
¹ Institute for Nanoelectronic devices and Quantum Computing, Fudan University, Shanghai 200433, China

² Department of Physics and State Key Laboratory of Surface Physics, Fudan University, Shanghai 200433, China

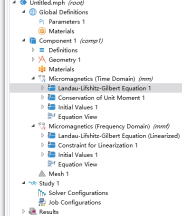
Introduction

Micromagnetic simulation is a widely-used technique to simulate dynamics of magnetic moments which is consistent with experimental observation. We build a Micromagnetics Module based on COMSOL by which one can not only perform time-domain micromagnetic simulation based on finite-element method, but also perform simulation in frequency-domain and obtain the capability of multi-physics coupling.

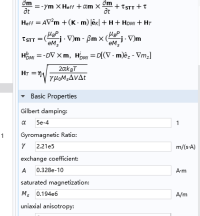
Select Physics



Model Builder



Settings



Time-domain Simulation

The dynamics of magnetic moments is governed by the Landau-Lifshitz-Gilbert (LLG) equation, where the effective fields include exchange interaction, anisotropy, Dzyaloshinskii-Moriya interaction (DMI), inter-layer coupling and thermal fluctuation etc. LLG equation is naturally time-dependent with nonlinearity, by solving which one can study excitation of spin waves as well as dynamics of magnetic textures.

$$\frac{\partial \mathbf{m}}{\partial t} = -\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}} + \alpha \mathbf{m} \times \frac{\partial \mathbf{m}}{\partial t}$$

$$\mathbf{H}_{\text{eff}} = A \nabla^2 \mathbf{m} + (\mathbf{K} \cdot \mathbf{m}) |\hat{\mathbf{e}}_K| + \mathbf{H} + \mathbf{H}_{\text{DMI}} + \mathbf{H}_T$$

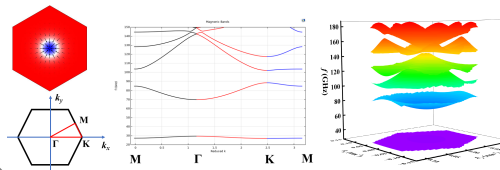


Frequency-domain Simulation

One unique feature of the module is the capability of solving LLG equation in frequency domain, so that one can directly calculate continuous excitation spectrums as well as discrete eigenmodes for a given magnetic texture. By applying Floquet boundary conditions, one can calculate magnonic energy bands of period structures such as skyrmion crystals.

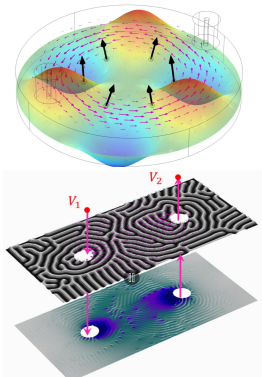
$$i\omega \delta \mathbf{m} = -\gamma \mathbf{m}_0 \times \delta \mathbf{h}^{\text{eff}} - \gamma \delta \mathbf{m} \times \mathbf{h}_0^{\text{eff}} + i\omega \alpha \mathbf{m}_0 \times \delta \mathbf{m}$$

$$\mathbf{m} = \mathbf{m}_0 + \delta \mathbf{m} e^{i\omega t} \quad \delta \mathbf{m}_{\text{dst}} = \delta \mathbf{m}_{\text{src}} e^{-ik_F \cdot (\mathbf{r}_{\text{dst}} - \mathbf{r}_{\text{src}})}$$



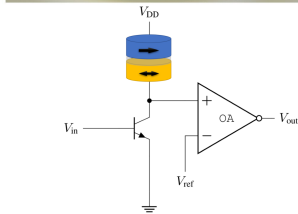
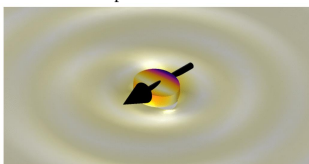
Capability of Multi-physics Coupling

Magnon-photon coupling for Cavity Spintronics



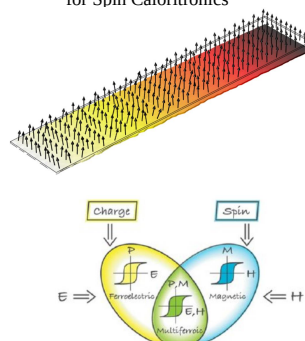
Magneto-resistive effects for Neuromorphic Spintronics

Magneto-elastic coupling for Spin Mechanics



Magnetic Tunnel Junction (MTJ) embedded in electric circuits for Probabilistic Computing

Thermal magnonic transport for Spin Caloritronics



Coupling between ferromagnetic moment and ferroelectric polarization for Multiferroics

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Tutorial



Intro to
Group Y



wcyu@fudan.edu.cn